

Persistence of Hourglass(-like) Structure: Improved Differential-Linear Distinguishers for Several ARX Ciphers

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Abstract. The ARX structure plays a crucial role in symmetric-key primitives, with differential-linear (DL) attacks being among the most effective cryptanalysis techniques against ARX ciphers. In this paper, we present a systematic re-decomposition technique for DL distinguishers of ARX ciphers and identify for the first time the hourglass(-like) structural commonalities among optimal DL distinguishers searched out by various deduction techniques, also supported through comprehensive experiments, which motivate us to develop an efficient and generalized approach to construct optimal hourglass(-like) structural DL distinguishers. Our method yields significant advances when applied to SPECK, Alzette, and the underlying permutations of SipHash and Chaskey: (1) the first 11- to 14-round DL distinguishers of Alzette; (2) the first (valid) DL distinguishers for 11-round SPECK32, 12-round SPECK48, and 16-round SPECK96; (3) *deterministic* (correlation ± 1) 3-round DL distinguishers for SipHash-2-4 and significantly improved 4-round ones. All these distinguishers are equipped with both theoretical and experimental verifications. We further analyze ARX-based Latin dance stream ciphers, achieving improved DL distinguishers for 7/7.25-round ChaCha, 8-round Salsa, and 5.5-round Forró. Though some of the improvements are not significant, we have verified the effectiveness of our method across a broader range of instances. This work provides new insights for DL distinguisher construction and enhances understanding of the security of ARX ciphers.

Keywords: ARX · Differential-linear distinguisher · Hourglass structure · SPECK · Alzette · Chaskey · SipHash · Latin dance stream ciphers

1 Introduction

Symmetric-key cryptographic primitives, including block ciphers, stream ciphers, hash functions and cryptographic permutations, etc., are considered as the workhorses of cryptography and play major roles in virtually any cryptographic scheme and any security-related application. Based on the selection of

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nonlinear components, we may divide most symmetric-key primitives into two categories namely the S-box based designs and the ARX designs. Different from the S-box based counterparts using word-level S-boxes as the source of non-linearity, the ARX designs, just as the name implies, only use three kinds of basic operations namely modular **A**dditions, word-wise **R**otations and **X**ORs, where modular additions are the only non-linear operation. Since the three basic operations are supported by nearly all hardware/software platforms, the ARX designs are not only efficient but also implementation-friendly, winning a high reputation in the realm of cryptography. Typical ARX primitives include block ciphers (LEA [26], HIGHT [27], SPECK [4], SPARX [21], etc.), stream ciphers (Salsa20 [11], ChaCha20 [10] etc.), hash functions (**S**kein [23], **B**LAKE [2], SipHash [1] etc.), the MAC algorithm Chaskey [35], also the cryptographic permutation Alzette [5].

Differential and linear attacks are the two powerful cryptanalysis methods for ARX designs. Analysis of S-box based designs usually happens at the word-level where secure bounds can be drawn by simply counting the minimum number of active S-boxes. On the contrary, differential/linear cryptanalysis of ARX designs can be harder because they have to be analyzed on a bit-level. Fortunately, many tools have been developed for deducing differential or linear trails of ARX ciphers, including automated tools based on general-purpose solvers (MILP, SAT or SMT) [22,24,36], and dedicated tools [12,13].

Differential-linear (DL) cryptanalysis is a technique proposed by Langford and Hellman at 1994 [29] that combines differential cryptanalysis with linear cryptanalysis such that their individual power over a small number of rounds can be joined to efficiently attack more rounds, which is proved effective for ARX primitives with applications to various ciphers such as Chaskey [30], ChaCha [7], and Salsa [15]. In DL attacks, a preliminary step is to search for DL distinguishers with high absolute correlations. The latest structure of DL distinguishers divides the targeted cipher E into three sub-ciphers, i.e., $E = E_2 \circ E_m \circ E_1$, where an r_1 -round differential characteristic (DC) covering E_1 , an r_2 -round linear characteristic (LC) covering E_2 , and an r_m -round DL approximation (DLA) covering E_m are deduced. Concatenating these three short distinguishers we obtain a long $(r_1+r_m+r_2)$ -round DL distinguisher under some independence assumptions.

Related work. At EUROCRYPT 2016, Leurent proposed a DL attack on Chaskey based on DL distinguishers found by a heuristic approach [30]. Beierle *et al.* later improved the general DL framework for ARX ciphers at CRYPTO 2020 [7], applying it to Chaskey and ChaCha. For ChaCha, they identified the first single-bit 3.5-round DLA, later formally verified by Dey *et al.* [18]. Coutinho *et al.* introduced Bidirectional Linear Expansions (BLE) at ASIACRYPT 2022 [15], enabling new DL attacks on ChaCha and Salsa. Their work yielded new 3.5-round DLAs for ChaCha, and the first 8-round DL distinguisher of Salsa based on a single-bit 5-round DLA, while also proposing the cipher Forró having the same security as ChaCha with fewer operations. Recently, Dey *et al.* [20] re-estimated the 5-round DL correlation of Salsa by introducing one more condition in the second round and obtained tighter correlation value of the 5-round DLA than the original one in [15]. While all previous DLAs of ChaCha

started with 1-bit input differences, Bellini *et al.* found new 3.5- and 4-round DLAs using 2-bit input differences [8], leading to better DL distinguishers than the previous. With the 4-round DLA from [8], [42] enhanced key recovery attacks against reduced-round ChaCha [42]. In the area of DL distinguishers searching, at CRYPTO 2022, Niu *et al.* developed an algorithm for theoretically evaluating the (rotational) DL correlations of ARX ciphers [37], resulting in qualified DL distinguishers against Alzette, SPECK32, and SipHash, while Bellini *et al.* [9] and Lv *et al.* [32] proposed MILP/MIQCP-based automated methods. At ASIACRYPT 2023, Chen *et al.* proposed a Meet-in-the-Middle (MITM) algorithm searching for good DLAs of ARX ciphers [14], built around the concept of strong unbalanced bit (SUB) and applied to DL attacks on SPECK and LEA.

Motivations. We thoroughly revisit the latest DL distinguisher deduction techniques on ARX ciphers as well as the resulting DL distinguishers [9,14,32,37], and find that they are either inaccurate or currently inefficient.

- In [37], assuming independence in each middle round of E_m may lead to a substantial loss of accuracy in correlation estimations particularly when dealing with a chain of modular additions.
- The automatic methods in [9,32] are largely limited by the model-solving process. Besides, they select DLA according to the theoretical correlation that heavily depends on round-wise independence assumptions within E_m , potentially resulting in accuracy losses and overlooking promising DLA candidates.
- [14] utilizes a heuristic search combined with experimental evaluation, which faces efficiency challenges due to the requirement of evaluating correlations across a large sample space for many multi-bit masks. Under current computational constraints, this methodology may potentially miss some highly qualified DL distinguishers. Furthermore, it lacks theoretical evaluations for the newly found DL distinguishers.

Our contributions. The *Hourglass* structure was first proposed to interpret DCs of Chaskey [35], where the difference is sparse in the middle of the DC (located in a single bit), and progressively become dense after the propagation in both backward and forward directions. In this paper, we argue that despite its traditional use in studying the DC properties of ARX designs, the hourglass(-like) structure still has potential to yield high-correlation DL distinguishers.

We first propose a systematic re-decomposition technique for DL distinguishers of ARX ciphers. Based on this, we revisit the latest DL distinguishers of SPECK, Alzette, LEA, and CHAM deduced by those mentioned above automatic or theoretical or experimental methods [9,14,32,37,39], and have identified for the first time the hourglass(-like) structural commonalities among the best DL distinguishers. Thus, we come to a heuristic conclusion supported by a comprehensive set of experiments on DL distinguishers of SPECK32 and Alzette that *Hourglass(-like) structural DL distinguishers tend to have higher absolute correlations, making them ideal construction targets*. We present the re-decomposition technique with applications to multiple ARX-based primitives, together with the support experiments on SPECK32 and Alzette in Section 3.

Building on the above insight, in Section 4 we develop an efficient and generalized approach by integrating multiple advanced techniques, including MILP-based search, SUB, BLE, and MITM techniques, to enhance DL distinguishers construction, where we propose new refined criteria and filtering strategies that effectively eliminate sub-optimal options while preserving advantageous ones, thereby significantly improving the overall performance. As applications, our new

Table 1: Comparisons of DL correlations of SPECK, Alzette, SipHash and Chaskey

Member	$\#R^\dagger$	$(r_1, r_m, r_2)^\dagger$	Estimation by $ bp\varepsilon^2 ^\ddagger$	Ref.
SPECK32	11	(3, 6, 2)	$2^{-16.68}$	[32]
	11	(3, 4, 4)	$2^{-16.19}$	[9]
	11	(3, 4, 4)	$2^{-15.83}$	Section 5.1
SPECK48	11	(2, 6, 3)	$2^{-17.40}$	[32, 14]
	12	(3, 6, 3)	$2^{-23.40}$	Section 5.1
SPECK64	11	(3, 5, 3)	$2^{-19.66}$	[32]
	11	(3, 5, 3)	$2^{-14.82}$	Section 5.1
	12	(4, 5, 3)	$2^{-23.72}$	[32]
	12	(3, 6, 3)	$2^{-22.29}$	Section 5.1
	13	(3, 7, 3)	$2^{-28.15}$	[14]
	13	(3, 7, 3)	$2^{-26.59}$	Section 5.1
SPECK96	15	(4, 7, 4)	$2^{-41.72}$	[14]
	15	(4, 7, 4)	$2^{-38.36}$	Section 5.1
	16	(4, 8, 4)	$*2^{-47.58}$	Section 5.1
Alzette	9	(2, 5, 2)	$2^{-7.60}$	[32]
	9	(2, 5, 2)	$2^{-7.27}$	Supp. Mat. J
	10	(2, 6, 2)	$2^{-10.52}$	[32]
	10	(2, 6, 2)	$2^{-10.53}$	Supp. Mat. J
	11	(2, 6, 3)	$2^{-14.53}$	Section 5.2
	12	(2, 7, 3)	$2^{-19.02}$	Section 5.2
	13	(3, 7, 3)	$2^{-24.32}$	Section 5.2
	14	(3, 7, 4)	$*2^{-31.26}$	Section 5.2
SipHash-2-4	3	—	$2^{-0.78}$	[37]
	3	(0, 2.5, 0.5)	1	Section 5.3
	4	—	$2^{-6.03}$	[37]
	4	(0, 3, 1)	$2^{-0.13}$	Section 5.3
Chaskey	7	(1.5, 4, 1.5)	$2^{-36.11}$	[7, 30]
	7	(1.5, 4, 1.5)	$2^{-33.65}$	Section 5.4

‡ b is experimentally computed while p and ε are derived using an MILP-based method.

$*$ The corresponding values of “ $|bp\varepsilon^2|$ ” are calculated with ε also experimentally derived.

† $\#R$ is the number of rounds distinguished by an (r_1, r_m, r_2) -hourglass(-like) structural DL distinguisher, i.e., an r_1 -round DC followed by an r_m -round DLA and then by an r_2 -round LC.

method can find all previous distinguishers with the underlying hourglass(-like) structure. Not only that, we apply it to a wide range of ARX-based primitives including SPECK family, Alzette, SipHash, and Chaskey, yielding new state-of-the-art DL distinguishers or improving upon existing ones, as shown in Section 5. The

comparisons are summarized in Table 1. We also provide the theoretical evaluations for these experimentally found DL distinguishers in Supp. Mat. N, together with theoretical interpretations for three previously reported experimental DL distinguishers of Chaskey without theoretical explanations ever before.

- For SPECK, we improve the DL distinguishers for 11-, 12- and 13-round SPECK64, and 15-round SPECK96 over the previous best ones given in [9,14,32,37], and propose the first (valid) 11-, 12-, and 16-round DL distinguishers for SPECK32, SPECK48, and SPECK96 respectively.
- For Alzette, we improve all existing DL distinguishers for different reduced-rounds Alzette and present the first 11- to 14-round ones.
- For the finalization of SipHash-2-4, we find many *deterministic* (correlation ± 1) 3-round DL distinguishers and significantly improve the correlation of the 4-round DL distinguisher from $2^{-6.03}$ to $2^{-0.13}$.
- For Chaskey, we improve the previous best 7-round DL distinguishers.

Furthermore, we apply our method to the well-known ARX-based stream ciphers of Latin dances in Section 6, improving the DL distinguishers over the previous. Comparisons are summarized in Table 2. For ChaCha, we improve the 7- and 7.25-round DL distinguishers based on the experimentally verifiable 3.5-round DLAs. Specifically, we demonstrate that the same “3.5DL+3.5LC” pattern starting from 1-bit input differences can still produce better DL distinguishers than 2-bit input differences, thereby challenging the claim made in [8]. For Salsa and Forró, we improve the previous best 8- and 5.25-round DL distinguishers respectively, and more importantly, we propose the first 5.5-round DL distinguishers for Forró. Though some of the improvements for Latin dance stream ciphers are not significant, we have verified the effectiveness of our method across a broader range of instances.

Table 2: Comparisons of DL distinguishers of ChaCha, Salsa and Forró

Cipher	# R	(r_1, r_m, r_2)	Complexity*	Ref.
ChaCha [†]	7	(0, 3.5, 3.5)	2^{214}	[15]
	7	(0, 3.5, 3.5)	2^{207}	[17]
	7	(0, 3.5, 3.5)	$2^{189.9}$	[8]
	7	(0, 3.5, 3.5)	$2^{184.52}$	Section 6.1
	7.25	(0, 3.5, 3.75)	$2^{260.14}$	[17]
	7.25	(0, 3.5, 3.75)	$2^{235.96}$	Section 6.1
Salsa	8	(0, 5, 3)	$2^{215.62}$	[15]
	8	(0, 5, 3)	$2^{192.1}$	Section 6.2
Forró	5.25	(0, 2, 3.25)	$2^{179.56}$	[15]
	5.25	(0, 2, 3.25)	$2^{177.64}$	Section 6.3
	5.5	(0, 2, 3.5)	2^{296}	[15]
	5.5	(0, 2, 3.5)	$2^{242.52}$	Section 6.3

[†] We compare only DL distinguishers also built on the experimentally verifiable 3.5-round DLAs (see explanation in Section 6.1), thus the 7-round DL distinguisher built on 4-round DLAs in [8] (with complexity $2^{166.89}$) was not in the comparison.

* We analyze the distinguishing attack complexity following the same methodology as in the compared paper, where data complexity equals time complexity ignoring the data limit.

The source code and supporting data of our experiments are available at <https://github.com/cryptoXinxinG/arxHourglass>

2 Preliminaries

2.1 Notations and Definitions

We use “ $\boxplus$ ” to denote the addition modulo 2^n (ADD) and “ $\oplus$ ” to denote the exclusive-OR. We denote an n -bit string $\mathbf{x}$ over $\mathbb{F}_2^n$ as $\mathbf{x} = (x_{n-1}, \dots, x_0)$, where x_{n-1} and x_0 are the most and least significant bits respectively. $|\mathbf{x}|$ denotes the length of $\mathbf{x}$. For two n -bit string $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^n$, the standard inner product is defined as $\mathbf{x} \cdot \mathbf{y} = \bigoplus_{i=0}^{n-1} x_i y_i$, where the product $x_i y_i$ is taken over $\mathbb{F}_2$. Let uppercase letters X, Y denote the uniformly distributed random variables for $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^n$. We denote the i -th unit vector of an n -bit binary vector by $[i]$, and the sum of unit vectors $[i_0] \oplus \dots \oplus [i_{m-1}]$ by $[i_{m-1}, \dots, i_0]$, where $n-1 \geq i_{m-1} > \dots > i_0 \geq 0$.

An n -variable Boolean function $f(\mathbf{x})$ is a mapping from $\mathbb{F}_2^n$ to $\mathbb{F}_2$. An (n, m) -function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is a mapping $\mathbf{x} \mapsto (f_0, \dots, f_{m-1})$, where f_i 's are n -variable Boolean functions. F is also called an m -dimensional vectorial Boolean function. For a binary random variable x , the correlation between x and zero is defined as $\varepsilon(x) = \Pr\{x = 0\} - \Pr\{x = 1\}$. Given a Boolean function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, the correlation of f to zero is defined as $\varepsilon(f) = \Pr\{f(X) = 0\} - \Pr\{f(X) = 1\}$. For the vectorial Boolean function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, we give the following definitions.

Definition 1. For input-output mask pair $(\boldsymbol{\lambda}, \boldsymbol{\lambda}') \in \mathbb{F}_2^n \times \mathbb{F}_2^m$, the correlation of the linear approximation $\boldsymbol{\lambda} \rightarrow \boldsymbol{\lambda}'$ of F is defined as $\varepsilon_F(\boldsymbol{\lambda} \rightarrow \boldsymbol{\lambda}') = \Pr\{\boldsymbol{\lambda}' \cdot F(X) = \boldsymbol{\lambda} \cdot X\} - \Pr\{\boldsymbol{\lambda}' \cdot F(X) \neq \boldsymbol{\lambda} \cdot X\}$. When $\varepsilon_F \neq 0$, we call $\boldsymbol{\lambda} \rightarrow \boldsymbol{\lambda}'$ an effective linear approximation of F , and define the correlation weight as $w_F^{(l)}(\boldsymbol{\lambda} \rightarrow \boldsymbol{\lambda}') = -\log |\varepsilon_F(\boldsymbol{\lambda} \rightarrow \boldsymbol{\lambda}')|$.

Definition 2. For input-output difference pair $(\boldsymbol{\delta}, \boldsymbol{\delta}') \in \mathbb{F}_2^n \times \mathbb{F}_2^m$, the differential probability of F is defined as $\Pr_F(\boldsymbol{\delta} \rightarrow \boldsymbol{\delta}') = \Pr\{F(X) \oplus F(X \oplus \boldsymbol{\delta}) = \boldsymbol{\delta}'\}$. When $\Pr_F \neq 0$, we call $\boldsymbol{\delta} \rightarrow \boldsymbol{\delta}'$ an effective difference propagation of F and define the weight of the difference propagation as $w_F^{(d)}(\boldsymbol{\delta} \rightarrow \boldsymbol{\delta}') = -\log \Pr_F(\boldsymbol{\delta} \rightarrow \boldsymbol{\delta}')$.

For an ARX primitive, we use $v_j[i_{m-1}, \dots, i_1, i_0]$ to denote either the n -bit linear mask $[i_{m-1}, \dots, i_1, i_0]$ or the n -bit difference $2^{i_0} + 2^{i_1} + \dots + 2^{i_{m-1}}$ at branch v_j , where $n-1 \geq i_{m-1} > \dots > i_1 > i_0 \geq 0$.

Definition 3. For the difference-mask pair $(\boldsymbol{\delta}, \boldsymbol{\lambda}) \in \mathbb{F}_2^n \times \mathbb{F}_2^m$, the correlation of the DLA $\boldsymbol{\delta} \rightarrow \boldsymbol{\lambda}$ of F is defined as $\mathcal{C}_F(\boldsymbol{\delta} \rightarrow \boldsymbol{\lambda}) = \Pr\{\boldsymbol{\lambda} \cdot (F(X) \oplus F(X \oplus \boldsymbol{\delta})) = 0\} - \Pr\{\boldsymbol{\lambda} \cdot (F(X) \oplus F(X \oplus \boldsymbol{\delta})) = 1\}$. When $\mathcal{C}_F \neq 0$, we call $\boldsymbol{\delta} \rightarrow \boldsymbol{\lambda}$ an effective DLA of F , and define the weight of the DLA as $w_F^{(a)}(\boldsymbol{\delta} \rightarrow \boldsymbol{\lambda}) = -\log |\mathcal{C}_F(\boldsymbol{\delta} \rightarrow \boldsymbol{\lambda})|$.

Definition 4. Let $A : \mathbb{F}_2^n \times \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be the modular addition operation such that $A(\mathbf{x}, \mathbf{y}) = \mathbf{x} \boxplus \mathbf{y}$. For the input difference $(\boldsymbol{\delta}_0, \boldsymbol{\delta}_1) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$ and the output mask $\boldsymbol{\lambda} \in \mathbb{F}_2^n$, let $G_{(\boldsymbol{\delta}_0, \boldsymbol{\delta}_1), \boldsymbol{\lambda}}(\mathbf{x}, \mathbf{y}) = \boldsymbol{\lambda} \cdot ((\mathbf{x} \boxplus \mathbf{y}) \oplus ((\mathbf{x} \oplus \boldsymbol{\delta}_0) \boxplus (\mathbf{y} \oplus \boldsymbol{\delta}_1)))$. Then the correlation of the DLA $(\boldsymbol{\delta}_0, \boldsymbol{\delta}_1) \rightarrow \boldsymbol{\lambda}$ of A can be computed as

$$\mathcal{C}_A((\boldsymbol{\delta}_0, \boldsymbol{\delta}_1) \rightarrow \boldsymbol{\lambda}) = \Pr\{G_{(\boldsymbol{\delta}_0, \boldsymbol{\delta}_1), \boldsymbol{\lambda}}(X, Y) = 0\} - \Pr\{G_{(\boldsymbol{\delta}_0, \boldsymbol{\delta}_1), \boldsymbol{\lambda}}(X, Y) = 1\}.$$

2.2 DL Attack

To ensure consistency with prior works [9,14,32,37], we adopt the latest structure of DL distinguishers in [3] for estimating DL correlations, as shown in Fig. 1. The cipher E of block size n is divided into three parts E_1 , E_m , and E_2 , s.t. $E = E_2 \circ E_m \circ E_1$, where DC and LC are applied to E_1 and E_2 successively. Suppose the r_1 -round DC $\delta' \rightarrow \delta$ holds with theoretical probability p , the r_2 -

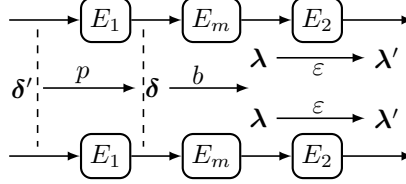


Fig. 1: The latest structure of DL distinguishers

round LC $\lambda \rightarrow \lambda'$ has a theoretical correlation ε , and the r_m -round DLA $\delta \rightarrow \lambda$ has the experimental correlation b , then the total correlation of the $(r_1 + r_m + r_2)$ -round DL distinguisher $\delta' \rightarrow \lambda'$ is estimated as $bp\varepsilon^2$.

Assume the attacker prepares $N = \alpha/(bp\varepsilon^2)^2$ pairs of chosen plaintexts $(x_i, x_i \oplus \delta')$, $i = 1, \dots, N$, where α is a small non-negative constant. When $N < 2^n$, where n is the block size of E , one can distinguish E from a pseudo-random permutation. We denote this DL distinguisher as

$$\delta' \xrightarrow{E_1} \delta \xrightarrow{E_m} \lambda \xrightarrow{E_2} \lambda', \text{ or } \delta' \xrightarrow[p]{r_1 \text{DC}} \delta \xrightarrow[b]{r_m \text{DL}} \lambda \xrightarrow[|\varepsilon|]{r_2 \text{LC}} \lambda',$$

and call it an (r_1, r_m, r_2) -DL distinguisher of E . Alternatively, we describe each of them as a DL distinguisher $\delta' \rightarrow \delta \rightarrow \lambda \rightarrow \lambda'$ following the (r_1, r_m, r_2) form.

2.3 Hourglass(-like) Structural DL Distinguishers

Hourglass structure. In [35], the designers of Chaskey stated that the best DCs they found for reduced-round Chaskey permutation have an “hourglass structure”: the difference is sparse in the middle of the DC (actually only a single bit is active), and this small difference is expanded by the avalanche effect when propagating in both directions. The same observation typically holds also for LCs. Due to these properties, given a good DL distinguisher with this hourglass structure, we can divide E into three parts $E = E_2 \circ E_m \circ E_1$, so that the single active bit in DC falls between E_1 and E_m and the single active bit in LC falls between E_m and E_2 . Then we use this decomposition to look for good hourglass structural DL distinguisher of E in the form of $\delta' \xrightarrow{\text{DC}} \delta \xrightarrow{\text{DL}} \lambda \xrightarrow{\text{LC}} \lambda'$, where δ is 1-bit input difference and λ is 1-bit output linear mask.

At EUROCRYPT 2016, Leurent presented the first cryptanalysis of Chaskey with a DL attack [30]. To find good DL distinguishers for Chaskey, the author

used a heuristic approach based on the above observation: most good DCs and good LCs have the hourglass structure, with a single active bit in the middle, and restrict himself to evaluate the middle DLA with a single active input bit and a single active output bit. A similar approach was utilized to search for DL distinguishers for Chaskey and ChaCha in [7], and for SPECK32 in [41].

Hourglass-like structure. The only non-linear operation for an ARX-based design is the modular addition. According to the propagation rules of LC over a modular addition as shown in Fig. 2, we argue that the LCs with two consecutive active bits in the middle round may also have high absolute linear correlations. For an output mask $[i]$ where $i \geq 2$, two LCs achieve the biggest

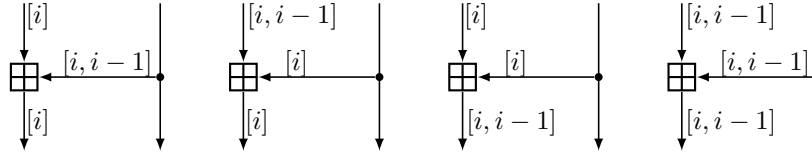


Fig. 2: The propagation rules of LC over a modular addition

absolute correlation of 2^{-1} : $([i], [i, i-1]) \rightarrow [i]$ and $([i, i-1], [i]) \rightarrow [i]$, characterized by the input mask pairs $([i], [i, i-1])$ and $([i, i-1], [i])$ respectively. Similarly, for an output mask $[i, i-1]$ where $i \geq 2$, two LCs $([i], [i]) \rightarrow [i, i-1]$ and $([i, i-1], [i, i-1]) \rightarrow [i, i-1]$ yield correlation 2^{-1} . Based on this, we argue that the optimal LCs may alternatively exhibit an “hourglass-like structure” with two consecutive active bits in some middle round. Similarly to the above, given a good DL distinguisher of $E = E_2 \circ E_m \circ E_1$ that essentially follows this hourglass-like structure, we can look for a good hourglass-like structural DL distinguisher where the position round with difference having only a single active bit lies between E_1 and E_m and that with the mask having two consecutive active bits lies between E_m and E_2 . Note that this matches both the DL distinguisher for Chaskey from [7] and our newly found DL distinguishers for 15-round SPECK96 and 11/12-round Alzette.

2.4 Existing Techniques

The SUB technique. In [14], Chen *et al.* introduced an MITM algorithm searching for good DLAs of ARX ciphers. Their approach leverages the observation that combining two DLAs with high absolute correlations will likely result in a new DLA with a relatively high correlation. The algorithm is built around the concept of *strong unbalanced bit* (SUB), a key principle we will refer to as the SUB technique throughout this paper.

For r_m -round DLAs, let ι be a given threshold of absolute correlations. For a preset difference δ , if the absolute experimental correlation of the r_m -round

DLA $\delta \rightarrow [j]$ evaluated under a given sample space exceeds ι , the j -th ciphertext bit is called a **strong unbalanced bit** (SUB). The set of all SUBs is denoted as $\mathcal{S}_\iota(\delta)$. Using the SUB technique, [14] designed an MITM algorithm searching for $\delta \xrightarrow{r_m \text{DL}} \sum \xrightarrow{r_2 \text{LC}} \lambda'$ over $E_2 \circ E_m$. Generally, they first prepare $\mathcal{S}_\iota(\delta)$ for given δ ; subsequently, they automatically search LCs $\lambda \xrightarrow{r_2 \text{LC}} \lambda'$ by adding constraints that $\lambda_i = 0, i \notin \mathcal{S}_\iota(\delta)$, and those on upper bound of the r_2 -round LC correlations or Hamming weights of λ' ; finally they experimentally evaluate $\delta \xrightarrow{r_m \text{DL}} \lambda$ for all those returned λ . We refer to [14] for details of the search algorithm. With this algorithm, improved DL distinguishers on SPECK and LEA are derived. We will show in Section 3 that all these distinguishers can be re-decomposed into the hourglass(-like) structure by using the re-decomposition technique.

The BLE technique. In [15], Coutinho *et al.* proposed the BLE technique exploring the expansion of a single bit in both forward and backward directions. In particular, the backward linear approximation is used to search for DLAs ending in the previous round. For example, suppose we can compute all possible single-bit DLAs for $(m-1)$ rounds of Salsa, and we have a backward LC (with correlation ± 1) from the m -th round to the $(m-1)$ -th round as $v_i[j] \rightarrow (v_{i_1}[s_1], \dots, v_{i_p}[s_p])$. Then we can use the Piling-up Lemma to aggregate the correlation for each single bit from the $(m-1)$ -th round, achieving a DL correlation for the m -round DLA ending with output mask $v_i[j]$ computed by multiplying DL correlations of those $(m-1)$ -round DLAs with output masks $v_{i_1}[s_1], \dots, v_{i_p}[s_p]$. As shown in [15], BLE is especially useful in situations where we do not have enough computational power to compute a DL correlation for the target single-bit output mask, but we can do so for each single-bit mask derived in backward direction individually. With the BLE technique, the first DL distinguishers ranging 7 and 8 rounds of Salsa are proposed. We refer to [15] for further details.

Remark 1. In Section 4, we propose a novel approach for identifying optimal hourglass(-like) structural DL distinguishers of ARX primitives, which integrates multiple advanced techniques, including the MILP-based search, SUB, BLE, and MITM techniques, to enhance DL distinguishers construction. Unlike the SUBs sets constructed in [14], which store only strong unbalanced bits, our SUBs sets additionally maintain the corresponding experimental correlations used for our effective filtering, as shown in Algorithm 5 and 6. Furthermore, while the BLE technique used in [15] estimates the DL correlation of an r_m -round DLA by aggregating the DL correlation for each single bit from the previous round based on the Piling-up Lemma, which may lead to an evaluation deviating from the actual correlation. Instead, we experimentally evaluate the r_m -round DL correlations during our search, ensuring more reliable hourglass(-like) structural distinguishers construction.

3 Re-decompositons to Hourglass(-like) Structural DL Distinguishers of ARX Ciphers

This section proposes a systematic re-decomposition technique for DL distinguishers of ARX ciphers. Applying it to DL distinguishers of SPECK, Alzette,

LEA, and CHAM [9,14,32,37,39], we identify the hourglass(-like) structural commonalities, leading to a heuristic conclusion that hourglass(-like) structural DL distinguishers tend to have higher absolute correlations, supported by intensive experiments on SPECK32 and Alzette. Due to space constraints, we provide the details of the involved ARX designs in Supp. Mat. A.

3.1 Re-decompositions of DL Distinguishers

As described in Section 2.3, if a good DL distinguisher $\delta' \xrightarrow{r_1 \text{DC}} \delta \xrightarrow{r_m \text{DL}} \lambda \xrightarrow{r_2 \text{LC}} \lambda'$ of $E = E_2 \circ E_m \circ E_1$ follows an hourglass(-like) structure, we may obtain an hourglass(-like) structural DL distinguisher of E such that the input difference of the middle DLA has a single active bit in some round between E_1 and E_m while the output mask of that has a single active bit or two consecutive active bits in some round between E_m and E_2 .

Given an (r_1, r_m, r_2) -DL distinguisher that essentially follows an hourglass(-like) structure, the position round where the difference with only a single active bit lies should be between E_1 and E_m , thus it can be one of the three cases: $< r_1$, $= r_1$, or $> r_1$. Similarly, the position round where the mask with a single active bit or two consecutive active bits lies should be between E_m and E_2 . Thus, one of the three cases could happen: $< r_1 + r_m$, $= r_1 + r_m$, or $> r_1 + r_m$. Accordingly, we can obtain an hourglass(-like) structural DL distinguisher following one of 9 possible forms. The next subsection will show how the best DL distinguishers for SPECK, Alzette, LEA, and CHAM all conform to one of these decompositions.

3.2 DL Distinguisher Re-decomposition: Speck, Alzette, LEA and Cham

Speck Family. For page limitation, we provide the latest DL distinguishers for reduced-round SPECK32/48/64/96/128 in Table 4 (Supp. Mat. B)[9,14,32,37,41]. All these DL distinguishers follow hourglass(-like) structures that we successfully re-decomposed, each following one of 9 forms discussed in Section 3.1.

Speck32. Table 5 (Supp. Mat. C) shows our hourglass(-like) structure re-decompositions of DL distinguishers of SPECK32 from Table 4. We focus on the 10-round cases from [9,14,32,37], noting that the 11-round DL distinguishers in [9,32] simply extend their 10-round counterparts backward by one round.

In [32], Lv *et al.* construct an $(r_1, r_m, r_2) = (5, 3, 2)$ -DL distinguisher with absolute experimental correlation $2^{-11.58}$, shown as

$$(0211, 0a04) \xrightarrow{E_1} (v_0[15], v_1[15, 10, 3, 1]) \xrightarrow{E_m} (v_1[1]) \xrightarrow{E_2} (5820, 4020).$$

By propagating the input difference/output mask of E_m bidirectionally, we locate the two position rounds where the difference with only a single active bit lies and the mask with a single active bit or two consecutive active bits lies. This yields a $(2, 6, 2)$ -hourglass re-decomposition of the $(5, 3, 2)$ -DL distinguisher with an estimated correlation $|bp\varepsilon^2| = 2^{-11.55}$:

$$(0211, 0a04) \xrightarrow[2^{-6}]{2\text{DC}} (v_0[6]) \xrightarrow[-2^{-3.55}]{6\text{DL}} (v_1[1]) \xrightarrow[2^{-1}]{2\text{LC}} (5820, 4020).$$

Additionally, we re-decompose two existing DL distinguishers into hourglass structures: (1) The (3, 3, 4)-DL distinguisher from [9] with absolute experimental correlation 2^{-12} : $(2050, 2040) \xrightarrow{E_1} (v_0[10], v_1[12, 10]) \xrightarrow{E_m} (v_1[3]) \xrightarrow{E_2} (3854, 3844)$, is re-decomposed to the (2, 4, 4)-hourglass structural DL distinguisher $(2050, 2040) \xrightarrow[2^{-3}]{2DC} (v_1[10]) \xrightarrow[-2^{-1.19}]{4DL} (v_1[3]) \xrightarrow[2^{-4}]{4LC} (3854, 3844)$ with absolute correlation estimated as $|bp\epsilon^2| = 2^{-12.19}$. (2) The (5, 4, 1)-DL distinguisher from [37] with absolute experimental correlation $2^{-13.9}$: $(0a20, 4205) \xrightarrow{E_1} (v_0[15, 8], v_1[15, 8, 1]) \xrightarrow{E_m} (v_0[3], v_1[3]) \xrightarrow{E_2} (5820, 4020)$, is re-decomposed to the (3, 5, 2)-hourglass structural DL distinguisher $(0a20, 4205) \xrightarrow[2^{-11}]{3DC} (v_0[6]) \xrightarrow[2^{-0.9}]{5DL} (v_1[1]) \xrightarrow[2^{-1}]{2LC} (5820, 4020)$ with absolute correlation $|bp\epsilon^2| = 2^{-13.9}$.

Additionally, Chen *et al.* [14] construct a 10-round DL distinguisher

$$(2800, 0010) \xrightarrow[2^{-2}]{1DC} (v_0[6]) \xrightarrow{6DL} \sum \xrightarrow{3LC} (2300, 4380),$$

where the 9-round DLA $(v_0[6]) \rightarrow (2300, 4380)$ of $E_2 \circ E_m$ is experimentally evaluated as $2^{-10.2}$, i.e., all the possible $\lambda \in \sum$ are considered, and thus the total correlation is estimated as $2^{-2} \times 2^{-10.2} = 2^{-12.2}$. We re-decompose it into the (1, 6, 3)-hourglass structure

$$(2800, 0010) \xrightarrow[2^{-2}]{1DC} (v_0[6]) \xrightarrow[-2^{-3.55}]{6DL} (v_1[1]) \xrightarrow[2^{-4}]{3LC} (2300, 4380)$$

with absolute correlation estimated as $|bp\epsilon^2| = 2^{-13.55}$. Note the deviation between two correlation values $2^{-12.2}$ and $2^{-13.55}$ stems from the clustering effects.

Speck48. Using an MILP/MIQCP-based automatic search method, Lv *et al.* [32] construct an 11-round $(r_1, r_m, r_2) = (6, 2, 3)$ -DL distinguisher for SPECK48:

$$(20082, 120200) \xrightarrow{E_1} (80a000, 85a420) \xrightarrow{E_m} (v_1[2]) \xrightarrow{E_2} (8a805, 98804)$$

with absolute theoretical/experimental correlation $2^{-20.46}/2^{-17.55}$. Similarly, we re-decompose it to the (2, 6, 3)-hourglass structural DL distinguisher

$$(20082, 120200) \xrightarrow[2^{-5}]{2DC} (v_0[15]) \xrightarrow[-2^{-6.40}]{6DL} (v_1[2]) \xrightarrow[2^{-3}]{3LC} (8a805, 98804), \quad (1)$$

with total correlation estimated as $|bp\epsilon^2| = 2^{-17.40}$. Notably, [14] also construct an 11-round DL distinguisher using the SUB technique:

$$(20082, 120200) \xrightarrow[2^{-5}]{2DC} (v_0[15]) \xrightarrow[-2^{-6.40}]{6DL} (v_1[2]) \xrightarrow[2^{-3}]{3LC} (c2801, d0800).$$

While maintaining the same hourglass structure, this distinguisher differs from Eq. (1) only in the output masks of the 3-round LC.

Speck64. Lv *et al.* [32] present an 11-round DL distinguisher for SPECK64 with absolute theoretical/experimental correlation $2^{-22.13}/2^{-19.47}$, and a 12-round one with absolute theoretical correlation $2^{-26.93}$ and no experimental results given (see the 1st and 2nd sub-rows of SPECK64 in Table 4). Similarly, we have re-

decomposed them to $(3, 5, 3)$ - and $(4, 5, 3)$ -hourglass structural DL distinguishers

$$\begin{aligned} (92400040, 40104200) &\xrightarrow[2^{-12}]{3\text{DC}} (v_0[7]) \xrightarrow[2^{-1.66}]{5\text{DL}} (v_1[5]) \xrightarrow[2^{-3}]{3\text{LC}} (4100400\text{c}, 4\text{c}004000), \\ (12029282, 02020282) &\xrightarrow[2^{-16}]{4\text{DC}} (v_1[21]) \xrightarrow[2^{-1.72}]{5\text{DL}} (v_1[5]) \xrightarrow[2^{-3}]{3\text{LC}} (61804028, 68004020), \end{aligned}$$

with total correlations estimated as $|bp\varepsilon^2| = 2^{-19.66}$ and $2^{-23.72}$ respectively.

Additionally, [14] presents the first 13-round DL distinguisher for SPECK64

$$(102490, 10801004) \xrightarrow[2^{-11}]{3\text{DC}} (v_0[21]) \xrightarrow[2^{-9.15}]{6\text{DL}} (v_0[8], v_1[29, 0]) \xrightarrow[2^{-4}]{4\text{LC}} (420\text{c}0200, 2400200)$$

with total correlation $|bp\varepsilon^2| = 2^{-28.15}$. Our $(3, 7, 3)$ -hourglass re-decomposition maintains the same correlation:

$$(102490, 10801004) \xrightarrow[2^{-11}]{3\text{DC}} (v_0[21]) \xrightarrow[2^{-9.15}]{7\text{DL}} (v_1[0]) \xrightarrow[2^{-4}]{3\text{LC}} (420\text{c}0200, 2400200).$$

Speck96. [14] presents the first 15-round DL distinguisher for SPECK96

$$\delta'_{opt} \xrightarrow[2^{-20}]{4\text{DC}} (v_0[45]) \xrightarrow[2^{-5.72}]{6\text{DL}} (v_0[21], v_1[13, 12, 10]) \xrightarrow[2^{-8}]{5\text{LC}} \lambda'_{opt}.$$

with correlation $|bp\varepsilon^2| = 2^{-41.72}$. Our $(4, 7, 4)$ -hourglass re-decomposition:

$$\delta'_{opt} \xrightarrow[2^{-20}]{4\text{DC}} (v_0[45]) \xrightarrow[2^{-5.75}]{7\text{DL}} (v_1[13]) \xrightarrow[2^{-7}]{4\text{LC}} \lambda'_{opt}$$

achieves higher correlation $|bp\varepsilon^2| = 2^{-39.75}$. Hereafter, δ'_{opt} and λ'_{opt} denote optimal input differences (for r_1 -round DC) and output masks (for r_2 -round LC), respectively, given their corresponding output differences and input masks.

Speck128. We can also re-decompose the 18-round DL distinguisher of SPECK128 constructed with $(r_1, r_m, r_2) = (5, 8, 5)$ in [14], to a $(5, 9, 4)$ -hourglass structural one. The detail is provided in Supp. Mat. E.

Alzette. Tables 6 and 7 (Supp. Mat. D) show that all DL distinguishers for reduced-round Alzette (up to 10 rounds) from [32, 37] — whether found automatically or theoretically — admit hourglass(-like) re-decompositions.

LEA. The first 16/17-round DL distinguishers of LEA [14] admits hourglass re-decompositions (Supp. Mat. E).

Cham. Building upon the MTIM search algorithm in [14], [39] presents a 46- and 58-round DL distinguisher of CHAM-64/128 and CHAM-128/256 respectively, admitting hourglass(-like) re-decompositions (Supp. Mat. E).

Heuristic Conclusion. From the above analysis, we have identified the hourglass(-like) structural commonalities among the best DL distinguishers on ARX ciphers, based on our systematic re-decomposition technique. This leads to our heuristic conclusion: *Hourglass(-like) structural DL distinguishers tend to exhibit higher absolute correlations than those based on other structures, making them ideal construction targets.* Intensive experiments on SPECK32 and Alzette (Section 3.3) strongly support this finding, motivating us to propose a new method to construct good DL distinguishers.

Remark 2. The exact correlation of a DL distinguisher $\delta' \rightarrow \lambda'$ is the sum of all DL trail correlations, requiring exhaustive enumeration of E_1 's output differences (δ) and E_2 's input masks (λ). Due to computational constraints, existing

methods typically consider only a single high-correlation DL trail, approximating the full DL distinguisher correlation by this dominant component. This approximation suffices as the deduced trail usually dominates the overall correlation with its significantly high absolute value.

3.3 The Support Experiments

We perform some experiments on SPECK32 and Alzette to further support our heuristic conclusion that hourglass(-like) structural DL distinguishers tend to exhibit higher absolute correlations. We list the experimental results on SPECK32 as summarized in Table 3, those on Alzette are given in Supp. Mat. F.

Table 3: Distributions of hourglass(-like) and non-hourglass(-like) structural DL distinguishers for SPECK32

$\#R(r_1, r_m, r_2)$	9-round (3, 4, 2)					10-round (3, 5, 2)				11-round (3, 4, 4)	
$-\log(bp\epsilon^2)$	[7,8)	[8,9)	[9,10)	[10,11)	[11,12)	[11,12)	[12,13)	[13,14)	[14,15)	[15,16)	[16,17)
#H(-like)	2	3	15	38	67	2	3	19	29	2	1
#non-H(-like)	0	0	0	3	18	0	0	0	1	0	0
#total	2	3	15	41	85	2	3	19	30	2	1
ratio of H(-like)	100%	100%	100%	93%	79%	100%	100%	100%	97%	100%	100%

Given $E = E_2 \circ E_m \circ E_1$, we construct DL distinguishers $\delta' \xrightarrow{E_1} \delta \xrightarrow{E_m} \lambda \xrightarrow{E_2} \lambda'$ by enumerating partial (δ, λ) pairs over E_m , due to the limitation on computational resources. We analyze the distributions of the hourglass(-like) and non-hourglass(-like) structural DL distinguishers across correlation ranges. We call (δ, λ) the m -to- n difference-mask pairs when δ and λ are of Hamming weight m and n respectively. In our experiments, the (δ, λ) pairs for exhaustion are limited to 1-to-1, 1-to-2, 2-to-1, 2-to-2, 1-to-3 and 3-to-1. Thus, there are a total of $\binom{32}{1} \cdot \binom{32}{1} + \binom{32}{1} \cdot \binom{32}{2} + \binom{32}{2} \cdot \binom{32}{1} + \binom{32}{2} \cdot \binom{32}{2} + \binom{32}{1} \cdot \binom{32}{3} + \binom{32}{3} \cdot \binom{32}{1} = 596,224$ (δ, λ) pairs to be exhausted for constructing DL distinguishers of SPECK32.

Searching Process. To construct ρ -round DL distinguishers of ARX primitives, we first identify r_1, r_2 , and r_m s.t. $r_1 + r_m + r_2 = \rho$. We search for DL distinguishers in a common three-stage way with the above difference-mask pairs: (1) search the optimal r_1 -round DCs and r_2 -round LCs for all 1-, 2- and 3-bit differences δ and masks λ , with theoretical probabilities and correlations denoted by p and ϵ respectively; (2) experimentally evaluate the r_m -round DLAs for all 596,224 (δ, λ) pairs, with the experimental correlation denoted by b ; (3) concatenate three short distinguishers into ρ -round DL distinguishers $\delta' \xrightarrow{r_1 \text{DC}} \delta \xrightarrow{r_m \text{DL}} \lambda \xrightarrow{r_2 \text{LC}} \lambda'$, and estimate the total correlation as $|bp\epsilon^2|$.

Experiment on 10-round Speck32. We specify $(r_1, r_m, r_2) = (3, 5, 2)$ to construct 10-round DL distinguishers and present in Table 3 the statistical results, where $x \in [a, b)$ denotes $a \leq x < b$, #H(-like) and #non-H(-like) count hourglass(-like) and non-hourglass(-like) structural DL distinguishers. The hourglass(-like) structural DL distinguishers obtained from our experiments consist of those that are themselves (3, 5, 2) hourglass(-like) structural ones, as well as those that can be re-decomposed into the hourglass(-like) structures following

the $(2, 6, 2)$, $(3, 4, 3)$, $(4, 4, 2)$, $(3, 6, 1)$, $(2, 5, 3)$, $(2, 7, 1)$, $(4, 5, 1)$ or $(4, 3, 3)$ forms. This yields 9 possible hourglass(-like) structural forms. For example, we obtain two 10-round DL distinguishers with absolute correlations $> 2^{-12}$:

$$\begin{aligned}\delta'_{opt} &\xrightarrow[2^{-6}]{3DC} (v_0[15], v_1[15]) \xrightarrow[-2^{-3.58}]{5DL} (v_1[1]) \xrightarrow[2^{-1}]{2LC} \lambda'_{opt}, \\ \delta'_{opt} &\xrightarrow[2^{-9}]{3DC} (v_0[3]) \xrightarrow[2^{-2.75}]{5DL} (v_1[5]) \xrightarrow[2^0]{2LC} \lambda'_{opt}.\end{aligned}$$

The first matches the hourglass re-decomposition of currently best 10-round DL distinguisher of SPECK32 in [32] when re-decomposed to $(2, 6, 2)$ -hourglass form:

$$\delta'_{opt} \xrightarrow[2^{-6}]{2DC} (v_0[6]) \xrightarrow[-2^{-3.58}]{6DL} v_1[1] \xrightarrow[2^{-1}]{2LC} \lambda'_{opt},$$

with $|bp\varepsilon^2| = 2^{-11.58}$. The second is new with $|bp\varepsilon^2| = 2^{-11.75}$, slightly worse than the currently best one. We also searched for 10-round DL distinguishers with $(r_1, r_m, r_2) = (2, 5, 3)$, but found none with absolute correlations $> 2^{-12}$. Notably, our experiments can reproduce all previous 10-round DL distinguishers from [9, 14, 32, 37]. Similarly, for 9-round distinguishers with $(r_1, r_m, r_2) = (3, 4, 2)$, results are summarized in Table 3.

Experiment on 11-round Speck32. Following the same process, we try to construct 11-round DL distinguishers. Note that under both settings of $(r_1, r_m, r_2) = (3, 4, 4)$ and $(2, 5, 4)$, we got three DL distinguishers with absolute correlations in the range $[15, 17)$, respectively, whose re-decompositions to hourglass(-like) structures are exactly the same shown as follows

$$\delta'_{opt} \xrightarrow[2^{-7}]{3DC} (v_1[10]) \xrightarrow[-2^{-1.19}]{4DL} (v_1[3]) \xrightarrow[2^{-4}]{4LC} \lambda'_{opt}, \text{ with } |bp\varepsilon^2| = 2^{-16.19} \quad (2)$$

$$\delta'_{opt} \xrightarrow[2^{-6}]{3DC} (v_1[3]) \xrightarrow[-2^{-1.89}]{4DL} (v_1[3]) \xrightarrow[2^{-4}]{4LC} \lambda'_{opt}, \text{ with } |bp\varepsilon^2| = 2^{-15.89} \quad (3)$$

$$\delta'_{opt} \xrightarrow[2^{-5}]{2DC} (v_0[1]) \xrightarrow[2^{-2.83}]{5DL} (v_1[3]) \xrightarrow[2^{-4}]{4LC} \lambda'_{opt}, \text{ with } |bp\varepsilon^2| = 2^{-15.83} \quad (4)$$

Similarly, in the ranges such as $[17, 18)$, the hourglass(-like) structural DL distinguishers dominate the absolute majority, but they are less interesting due to low absolute correlation values. We give results for the interesting ranges $[15, 16)$ and $[16, 17)$ in the right-most part of Table 3. Notably, DL distinguisher in Eq. (2) matches the re-decomposition to hourglass structure of the previous best one in [9]³, also as shown in the second row from the bottom of Table 5, and DL distinguishers in Eq. (3) and Eq. (4) are our new and improved 11-round ones, which are also the first two DL distinguishers for 11-round SPECK32 with absolute correlations strictly higher than 2^{-16} .

4 New DL Distinguishers Searching Method

Building on the heuristic conclusion in Section 3, we develop a component-wise approach to construct optimal hourglass(-like) structural DL distinguishers.

³ The total correlation of $-2^{-16.00}$ in [9] stems from DL clustering effects.

4.1 Description of Our Algorithm

We divide the target primitive E as usual. We will search good DCs for E_1 and LCs for E_2 using mature MILP-based automatic methods [24], while selecting good DLAs for E_m by experimentally evaluating their correlations.

While exhaustively verifying all difference-mask pairs over E_m is impractical, we argue that effective DL distinguishers can be derived based on an hourglass(-like) structure with 1-bit input differences and 1-bit or consecutive 2-bit output masks for E_m . For constructing ρ -round DL distinguishers of ARX primitives, we first specify the decomposition parameters (r_1, r_m, r_2) with $r_1 + r_m + r_2 = \rho$. Our goal is to find 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$, where $\delta = [i]$ and $\lambda = [j]$ or $[j+1, j]$, forming a high-correlation DL distinguisher $\delta'_{opt} \xrightarrow{r_1 \text{DC}} \delta \xrightarrow{r_m \text{DL}} \lambda \xrightarrow{r_2 \text{LC}} \lambda'_{opt}$.

DC part: For an r_1 -round DC $\delta' \xrightarrow{r_1 \text{DC}} \delta$, we denote the theoretical differential probability by p and its weight by $w_d = -\log(p)$. For each $\delta = [i]$, the optimal r_1 -round DCs are searched out by an automatic model $\mathbf{Model}_{\text{DC}}()$.

LC part: For an r_2 -round LC $\lambda \xrightarrow{r_2 \text{LC}} \lambda'$, we denote the theoretical absolute correlation by $|\varepsilon|$ and its weight by $w_l = -\log(|\varepsilon|)$. For each $\lambda = [j]$ or $[j+1, j]$, the optimal r_2 -round LCs are searched out by an automatic model $\mathbf{Model}_{\text{LC}}()$.

DLA part: For an r_m -round DLA $\delta \xrightarrow{r_m \text{DL}} \lambda$, we experimentally evaluate the correlation $|b|$ for each candidate difference-mask pair (δ, λ) filtered by our proposed strategy, where $\delta = [i]$ and $\lambda = [j]$ and $[j+1, j]$. We highlight that the choice of r_m is critical to our construction:

- **Small r_m case:** When r_m is relatively small, the values of $|b|$ tend to be somewhat large for most of the above (δ, λ) pairs. This allows for either reliable estimation of $|b|$ through experiments with a limited number of pseudociphertext pairs, or efficient filtering of insignificant pairs.
- **Large r_m case:** As r_m increases, the absolute correlations decreases rapidly, most (δ, λ) pairs typically exhibit considerably smaller $|b|$ values. This scenario demands a significantly larger sample space for accurate correlation evaluation. Because of that, the computation quickly becomes infeasible.

Algorithm 1: General description of the component-wise algorithm

Input: Block size n ; The values of r_1 , r_2 and r_m .

Output: $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ with the highest absolute DL correlation.

- 1 Run Algorithm 2 to prepare tables $\mathcal{T}^{wt_0}, \dots, \mathcal{T}^{wt_{ct-1}}$, $wt_0 < \dots < wt_{ct-1}$;
 - 2 **if** $r_m \leq cr$ **then**
 - 3 Run Algorithm 3 to obtain good 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$;
 - 4 **else**
 - 5 Run Algorithm 5 or 6 to identify SUBs sets for all 1-bit input differences;
 - 6 Run Algorithm 4 to obtain good 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$;
-

We formally introduce the concept of *critical-round number* (denoted as cr) for DLAs. This crucial parameter represents the maximum number of middle rounds r_m for which the absolute correlations $|b|$ under most (δ, λ) pairs with

$\delta = [i]$ and $\lambda = [j]$ (resp. $[j + 1, j]$) can be reliably estimated within practical computational limits. cr is an important threshold that naturally partitions our search space into two cases: $r_m \leq cr$ and $r_m > cr$, which are handled separately.

A high-level description of the overall search process is presented in Algorithm 1, following a component-wise approach. In line 5 of Algorithm 1, either Algorithm 5 or 6 is employed to identify the SUBs sets. As demonstrated in Section 4.3, we introduce Algorithm 5, which utilizes byte/half-word-based masks over extension fields to enhance the efficiency for constructing the SUBs set compared to Algorithm 6. While we recommend Algorithm 5 for its improved efficiency, we retain Algorithm 6 to ensure reproducibility and maintain consistency with prior work from [14].

As an empirical value, cr is chosen according to specific ARX primitives and the preset sample size for experimental evaluations. For example, in the case of Alzette (block size $n = 64$), we experimentally evaluate $64 \times 126 = 2^{12.98}$ r_m -round DLAs of the above form using $N (= 2^{30})$ ciphertext pairs. A simple code is used to test various r_m values (e.g. 3, 5, ...), adjusting it according to the experimental correlations of the r_m -round DLAs obtained. Results show that DLAs with up to 6 rounds yield accurate estimates, while longer ones do not. Thus, we set $cr = 6$ for Alzette. Our experiments further reveal that as the block size n of the ARX primitive increases, the absolute correlation $|b|$ tends to be higher for the same value of cr . For instance, while $cr = 6$ is suitable for SPECK64, SPECK96, and Alzette, the values differ for other variants: $cr = 5$ for SPECK48 and $cr = 7$ for SPECK128.

Remark 3. We present an efficient parameter selection strategy for determining the optimal values of cr , r_1 , r_2 , and r_m . While an exhaustive search over all decompositions satisfying $r_1 + r_m + r_2 = \rho$ would be theoretically possible, such an approach proves computationally inefficient. Instead, we implement a more practical three-step method: First, determine the critical-round number cr . Then, set r_m near cr (typically $cr \pm 1$ or $cr \pm 2$). Finally, for fixed r_m , select r_1 and r_2 to minimize the weight function $wt = -\log(p) - 2\log(|\varepsilon|)$ (computed via Algorithm 2), as the total DL distinguisher correlation is estimated by $bp\varepsilon^2$.

4.2 Component-wise Searching Strategy

In this section, we demonstrate our search algorithm step by step.

Firstly, we construct the MILP-based automatic models **Model**_{DC}() and **Model**_{LC}() for searching DCs and LCs of ARX ciphers following [24], based on which we deal with the DC and LC parts using Algorithm 2.

- Line 2-5 of Algorithm 2 prepare table $\mathcal{T}_{DC}$ storing n optimal DC pairs (δ'_{opt}, δ) where $\delta = [i], i = 0, \dots, n - 1$. We execute **Model**_{DC}() modeling ARX operations (branch/XOR/rotation/modular addition), and optimize the r_1 -round DC weight w_d . These pairs are stored by ascending w_d in $\mathcal{T}_{DC}$.
- Line 6-9 prepare table $\mathcal{T}_{LC}$ storing $\approx 2n - 1$ optimal LC pairs $(\lambda, \lambda'_{opt})$ for $\lambda = [j]$ and $[j + 1, j]$, searched out using the MILP-based tool by optimizing the r_2 -round LC weight w_l . These pairs are stored by ascending w_l in $\mathcal{T}_{LC}$.

Algorithm 2: Automatic search of $\delta'_{opt} \xrightarrow{r_1 \text{DC}} \delta$ and $\lambda \xrightarrow{r_2 \text{LC}} \lambda'_{opt}$

Input: Block size n ; The values of r_1 and r_2 .
Output: Tables $\mathcal{T}^{wt_0}, \mathcal{T}^{wt_1}, \dots, \mathcal{T}^{wt_{ct-1}}$, $wt_0 < wt_1 < \dots < wt_{ct-1}$.
1 $\mathcal{T}_{DC} \leftarrow []$; $\mathcal{T}_{LC} \leftarrow []$;
2 **for** $i \in \{0, \dots, n-1\}$ **do**
3 $\delta \leftarrow [i]$;
4 Search $\delta'_{opt} \xrightarrow{r_1 \text{DC}} \delta$ by running **Model**_{DC}() , compute $w_d = -\log(p)$;
5 Store (δ'_{opt}, δ) in a row of $\mathcal{T}_{DC}$ indexed by w_d in ascending order;
6 **for** $j \in \{0, \dots, n-1\}$ **do**
7 $\lambda \leftarrow [j]$ (resp. $[j+1, j]$);
8 Search $\lambda \xrightarrow{r_2 \text{LC}} \lambda'_{opt}$ by running **Model**_{LC}() , compute $w_l = -\log(|\varepsilon|)$;
9 Store $(\lambda, \lambda'_{opt})$ in a row of $\mathcal{T}_{LC}$ indexed by w_l in ascending order;
10 Store $\approx 2n^2$ 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ in $\mathcal{T}^{wt}$ according to $wt = w_d + 2w_l$

- In line 10, about $2n^2$ 4-tuples are classified into separate tables $\mathcal{T}^{wt_0}, \mathcal{T}^{wt_1}, \dots, \mathcal{T}^{wt_{ct-1}}$ based on their $wt = w_d + 2w_l$ values, where ct is the distinct wt count and $wt_0 < wt_1 < \dots < wt_{ct-1}$. The lowest- wt tuples are in $\mathcal{T}^{wt_0}$, and the highest in $\mathcal{T}^{wt_{ct-1}}$. These ct tables will feed into Algorithm 3 or Algorithm 4.

Secondly, we experimentally evaluate r_m -round DLAs for those candidate difference-mask pairs (δ, λ) using sufficient pseudo-ciphertext pairs, considering two cases: $r_m \leq cr$ and $r_m > cr$, within computational constraints.

Case 1 ($r_m \leq cr$). We will experimentally evaluate the r_m -round DLAs for $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ in tables $\mathcal{T}^{wt_0}, \dots, \mathcal{T}^{wt_{ct-1}}$ while filtering:

Algorithm 3: Search of $\delta'_{opt} \xrightarrow{r_1 \text{DC}} \delta \xrightarrow{r_m \text{DL}} \lambda \xrightarrow{r_2 \text{LC}} \lambda'_{opt}$ when $r_m \leq cr$

Input: Tables $\mathcal{T}^{wt_0}, \dots, \mathcal{T}^{wt_{ct-1}}$; Block size n ; The value of r_m ; Sample size N .
Output: $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ with highest absolute correlation $2^{-\pi(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})}$.
// Prepare the pseudo-ciphertext pairs for E_m
1 **for** $l = 1, \dots, N$ **do**
2 Randomly generate the plaintext $\mathbf{x}_l$;
3 Compute $\mathbf{y}_l = E_m(\mathbf{x}_l)$, and $\mathbf{y}_l^{[i]} = E_m(\mathbf{x}_l \oplus [i])$ for $i = 0, \dots, n-1$;
// Filtering-while-experimentally evaluating
4 $k \leftarrow 0$; $u \leftarrow ct$; $\tau \leftarrow +\infty$;
5 **while** $k < u$ **do**
6 **for** each $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ in $\mathcal{T}^{wt_k}$ **do**
7 Compute DL correlation $|b|$ of $\delta \xrightarrow{r_m \text{DL}} \lambda$ over N pairs $(\mathbf{y}_l, \mathbf{y}_l^\delta)$;
8 Compute the correlation weight $\pi(\delta'_{opt}, \delta, \lambda, \lambda'_{opt}) = wt_k - \log(|b|)$;
9 $\tau \leftarrow \min\{\min_{(\delta'_{opt}, \delta, \lambda, \lambda'_{opt}) \in \mathcal{T}^{wt_k}} \{\pi(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})\}, \tau\}$;
10 $u \leftarrow \min_{c=k+1}^{u-1} \{c : wt_c \geq \tau\}$;
11 $k \leftarrow k + 1$;

- Line 1-3 of Algorithm 3 prepare the pseudo-ciphertext pairs, used for experimental verifications of the $r_m(\leq cr)$ -round DLAs.

- Line 4–11 implement an efficient *filtering-while-experimentally evaluating* strategy to rapidly identify 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ exhibiting the highest absolute DL correlations. Specifically, we first experimentally evaluate the r_m -round DLAs for each 4-tuple in $\mathcal{T}^{wt_0}$ and record the maximum observed absolute correlation value with weight denoted as τ . Subsequently, all 4-tuples in $\mathcal{T}^{wt_c}$ satisfying $wt_c \geq \tau$ are filtered out, i.e., excluded from further consideration in the search process. This procedure is repeated iteratively, progressively narrowing down the search space.

Following the above, we hope to significantly accelerate the overall search for $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ with the highest absolute DL correlation $2^{-\pi(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})}$.

Complexity Estimation: Compared to the baseline approach with the complexity $\mathcal{O}(2n^2N)$, our *filtering-while-experimentally evaluating* strategy reduces the complexity to $\mathcal{O}(X \cdot N)$ with X being the number of filtered candidates.

Algorithm 4: Search of $\delta'_{opt} \xrightarrow{r_1 \text{DC}} \delta \xrightarrow{r_m \text{DL}} \lambda \xrightarrow{r_2 \text{LC}} \lambda'_{opt}$ when $r_m > cr$

Input: Threshold parameters (ι, χ) ; Tables $\mathcal{T}^{wt_0}, \dots, \mathcal{T}^{wt_{ct-1}}$; SUBs sets $\mathcal{S}_\iota([i])$ of $(r_m - t)$ -round DLAs for all $i = 0, \dots, n - 1$; Sample size N .
Output: $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ with highest absolute correlation $2^{-\pi(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})}$.

// Prepare pseudo-ciphertext pairs for E_m

- 1 **for** $l = 1, \dots, N$ **do**
- 2 Randomly generate the plaintext $\mathbf{x}_l$ and compute $\mathbf{y}_l = E_m(\mathbf{x}_l)$;
- 3 Compute $\mathbf{y}_l^{[i]} = E_m(\mathbf{x}_l \oplus [i])$ for $i = 0, \dots, n - 1$;
- // Filtering-while-experimentally evaluating
- 4 $k \leftarrow 0$; $u \leftarrow ct$; $\tau \leftarrow +\infty$;
- 5 **while** $k < u$ **do**
- 6 $\mathcal{C} \leftarrow []$;
- 7 **for each** $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ in $\mathcal{T}^{wt_k}$ **do**
- 8 **if** δ and λ collide with approximate weight $\psi(\delta, \lambda)$ under (ι, χ) **then**
- 9 **if** $(wt_k + \psi(\delta, \lambda)) \leq \tau$ **then**
- 10 Add $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ to $\mathcal{C}$;
- 11 **if** $\mathcal{C} \neq \emptyset$ **then**
- 12 **for each** $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt}) \in \mathcal{C}$ **do**
- 13 Compute DL correlation $|b|$ of $\delta \xrightarrow{r_m \text{DL}} \lambda$ over N pairs $(\mathbf{y}_l, \mathbf{y}_l^{[i]})$;
- 14 Compute the correlation weight $\pi(\delta'_{opt}, \delta, \lambda, \lambda'_{opt}) = wt_k - \log(|b|)$;
- 15 $\tau \leftarrow \min\{\min_{(\delta'_{opt}, \delta, \lambda, \lambda'_{opt}) \in \mathcal{C}} \{\pi(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})\}, \tau\}$;
- 16 $u \leftarrow \min_{c=k+1}^{u-1} \{c : wt_c \geq \tau\}$;
- 17 $k \leftarrow k + 1$;

Case 2 ($r_m > cr$). In this case, we do not have enough computational power to evaluate numerous r_m -round DLAs experimentally, but we can do so for $(r_m - t)$ -round DLAs, where t is a small positive integer. Our search strategy integrates the BLE and SUB techniques (see Remark 1) through an MITM approach over E_m , employing an enhanced filtering criterion to efficiently eliminate sub-optimal 4-tuples while preserving promising candidates. Specifically, for r_m -round DLAs, we apply the SUB technique (forward) for the first $(r_m - t)$ rounds and the BLE

technique (backward) for the remaining t rounds. This hybrid approach yields significant performance gains in the search process.

For a given threshold ι and an input difference $\delta = [i]$, let b_j^i denote the experimental correlation of the $(r_m - t)$ -round DLA $[i] \rightarrow [j]$. The SUBs set $\mathcal{S}_\iota([i])$ is then constructed by including all pairs $(j, |b_j^i|)$ satisfying $|b_j^i| > \iota$. The complete procedure for generating $\mathcal{S}_\iota([i])$ for all 1-bit input differences is formalized in Algorithm 6 (Supp. Mat. G). For mask $\hat{\lambda} = [j_1, \dots, j_k]$, we write $\hat{\lambda} \subset \mathcal{S}_\iota([i])$ to indicate that all active bits j_θ , $\theta = 1, \dots, k$, of $\hat{\lambda}$ are contained in $\mathcal{S}_\iota([i])$.

To illustrate our search strategy, we introduce two key concepts related to the (δ, λ) pair over E_m : “collide” and “approximate weight”. Let χ be a predefined threshold for absolute correlations of t -round LCs. Using the precomputed SUBs sets $\mathcal{S}_\iota([i])$ for $(r_m - t)$ -round DLAs, we say that $\delta = [i]$ and λ *collide* if there exists $\hat{\lambda} (= [j_1, \dots, j_k]) \subset \mathcal{S}_\iota([i])$ such that the t -round LC $\hat{\lambda} \rightarrow \lambda$ has correlation $\hat{\epsilon}$ satisfying $|\hat{\epsilon}| \geq \chi$, and if so we define $\phi([i], \hat{\lambda}) = \max_{\theta=1}^k \{-\log |b_{j_\theta}^i|\}$ for such masks. Among all colliding multi-bit masks, we define the *approximate weight* of the r_m -round DL correlation between δ and λ as: $\psi([i], \lambda) = \min_{\hat{\lambda} \subset \mathcal{S}_\iota([i])} \{\phi([i], \hat{\lambda})\}$. It is worth noting that the approximate weight provides a lower bound for the weight of the middle r_m -round DL correlation under (δ, λ) . This bound is derived by analyzing the colliding multi-bit masks between δ and λ , leveraging the pre-computed SUBs sets for $(r_m - t)$ -round DLAs. We give an example about these two concepts in Supp. Mat. H.

Based on the above, we present Algorithm 4 for finding optimal DL distinguishers $\delta'_{opt} \xrightarrow{r_1 \text{DC}} \delta \xrightarrow{r_m \text{DL}} \lambda \xrightarrow{r_2 \text{LC}} \lambda'_{opt}$ when $r_m > cr$. Similarly, lines 1-3 prepare pseudo-ciphertext pairs for verifying r_m -round DLA candidates. During the *filtering-while-experimentally evaluating* stage, we process 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ from tables $\mathcal{T}^{wt_0}, \mathcal{T}^{wt_1}, \dots, \mathcal{T}^{wt_{ct-1}}$ sequentially:

- Line 8 handles the two aforementioned concepts. The collision search between $\delta = [i]$ and $\lambda = [j]$ (or $[j+1, j]$) over E_m resembles Stage 2 of Algorithm 2 in [14] for $E_2 \circ E_m$. Concretely, for current (δ, λ) pair, we automatically search LCs $\hat{\lambda} \xrightarrow{t \text{LC}} \lambda$ (correlation $\hat{\epsilon}$) with constraints that $\hat{\lambda}_i = 0$, $i \notin \mathcal{S}_\iota(\delta)$, and $|\hat{\epsilon}| \geq \chi$. The approximate weight can be derived from the colliding $\hat{\lambda}$ masks via $\mathcal{S}_\iota(\delta)$.
- Line 9-10 combine the approximate weight with the DC probability and LC correlation weights to compute a lower bound for the weight of the total DL correlation of the 4-tuple $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$. The value of the threshold parameter τ is dynamically optimized through iterative experimental validation, and candidates are selected based on these bounds and the current value of τ .
- Line 12-15 experimentally evaluate the r_m -round DLAs under sufficient pseudo-ciphertext pairs and yield the total DL correlations for the 4-tuple candidates. Parameters τ and u are dynamically updated, progressively narrowing the search based on the observed maximum absolute DL correlations.

Complexity Estimation: We apply a two-phase evaluation strategy. First, we prepare SUBs sets for $(r_m - t)$ -round DLAs by evaluating n^2 single-bit difference-mask pairs over N' samples, then we evaluate X filtered candidates over N samples. Generally, our method reduces the complexity from $\mathcal{O}(2n^2N)$ to $\mathcal{O}(n^2N' + XN)$.

Remark 4. The parameters (t, ι, χ, N') are determined empirically based on specific ARX primitives, which are interrelated and chosen through iterative testing and adjustment, taking into account factors such as experimental correlations, t -round LC correlations, etc. We refer to Sections 5 and 6 for their values applied to SPECK, ChaCha, etc.

Comparison with [14]. While incorporating the SUB technique from [14], our approach differs in several key aspects:

1. For $r_m \leq cr$, our approach replaces the computationally intensive method of [14] preparing SUBs sets for all single-bit difference with an efficient “filtering-while-experimentally evaluating” strategy.
2. For $r_m > cr$, our approach shows superior efficiency in searching $(r_1 + r_m + r_2)$ -round DL distinguishers through two advantages: (1) a global search strategy and (2) effective filtering based on approximate weight. In contrast, [14] requires preparing SUBs sets of $r_m (> cr)$ -round DLAs for all single-bit differences and subsequent experimentally evaluating the DL correlations for all colliding multi-bit masks, leading to significantly higher computational cost.
3. For $r_m > cr$, the collision search between $\delta = [i]$ and $\lambda = [j]$ (or $[j+1, j]$) over E_m resembles Stage 2 of Algorithm 2 in [14] targeting at $E_2 \circ E_m$. If adapting to their approach, in their framework, E_m should be reinterpreted to cover $(r_m - t)$ rounds with input difference $\delta = [i]$ and E_2 covers t rounds with output mask $\lambda = [j]$ (or $[j+1, j]$). While Algorithm 2 in [14] could theoretically replace our r_m -round DLA search, this would introduce significant inefficiencies:
 - **Computational Cost:** It requires experimentally evaluating $(r_m - t)$ -round DL correlations for all single-bit input differences and colliding multi-bit output masks, which is time-consuming. Our method avoids this by leveraging pre-computed $(r_m - t)$ -round SUBs sets via the concept of approximate weight, drastically reducing computation.
 - **Holistic Search:** Even if high-quality r_m -round DLAs are found using [14], one must revert to considering E_1 and E_2 separately. Our approach remains superior as it jointly searches E_1 , E_m , and E_2 , ensuring globally optimal DL distinguishers.

4.3 A Trick for Improving the Efficiency

For a given difference $\delta (= [i])$, a straightforward approach for preparing the SUBs set is to experimentally evaluate $\delta \rightarrow [j]$ for each single-bit mask $\lambda = [j]$ over N' pseudo-ciphertext pairs $(\mathbf{y}_l, \mathbf{y}_l^\delta)$, as shown in line 6-9 of Algorithm 6. Thus the time complexity is $\mathcal{O}(nN')$. Here we use a small trick for reducing the time consumption: Instead of evaluating the correlation for each binary linear mask $[j]$ over $\mathbb{F}_2$, we evaluate that for the byte-based or half-word-based linear masks over $\mathbb{F}_{2^8}$ or $\mathbb{F}_{2^{16}}$, the extension fields of $\mathbb{F}_2$.

To illustrate the idea, we take the case where block size $n = 64$ as an example, which can be adapted to common cases like $n = 32, 48, 64, 96, 128, \dots$. For the given parameters r_m and t , we first prepare 64-bit pseudo-ciphertext pairs $(\mathbf{y}_l, \mathbf{y}_l^{[i]})$ of

$(r_m - t)$ -round DLAs for $i = 0, \dots, 63$ and $l = 1, \dots, N'$. Each $(\mathbf{y}_l, \mathbf{y}_l^{[i]})$ can be written into four 16-bit blocks as $\mathbf{y}_l = (\mathbf{y}_{l,0} \parallel \mathbf{y}_{l,1} \parallel \mathbf{y}_{l,2} \parallel \mathbf{y}_{l,3})$ and $\mathbf{y}_l^{[i]} = (\mathbf{y}_{l,0}^{[i]} \parallel \mathbf{y}_{l,1}^{[i]} \parallel \mathbf{y}_{l,2}^{[i]} \parallel \mathbf{y}_{l,3}^{[i]})$. In Algorithm 5, we present the detailed process on how to reduce the time consumption of preparing SUBs sets by considering half-word-based masks, by which the time complexity for preparing the SUBs set for a given difference is reduced from $\mathcal{O}(64N')$ to $\mathcal{O}(4(N' + 2^{20}))$, approximately an improvement of a factor of 2^4 .

Algorithm 5: Identify SUBs sets for all 1-bit input differences ($n = 64$)

Input: A threshold ι ; Sample size N' .
Output: SUBs sets $\mathcal{S}_\iota([i])$ for all $i = 0, \dots, 63$.

- 1 Prepare N' 64-bit pseudo-ciphertext pairs $(\mathbf{y}_l, \mathbf{y}_l^{[i]})$ for $i = 0, \dots, 63$;
- 2 Prepare an empty array $\mathcal{D}$ of size 2^{16} ;
- 3 **for** $i = 0, \dots, n - 1$ **do**
- 4 $\mathcal{S}_\iota([i]) \leftarrow []$;
- 5 **for** $k = 0, 1, 2, 3$ **do**
- 6 $\mathcal{D} \leftarrow \mathbf{0}$;
- 7 **for** $l = 1, \dots, N'$ **do**
- 8 Compute the 16-bit $v \triangleq \mathbf{y}_{l,k} \oplus \mathbf{y}_{l,k}^{[i]}$;
- 9 $\mathcal{D}[v] \leftarrow \mathcal{D}[v] + \frac{1}{N'}$;
- 10 **for** $j = 0, \dots, 15$ **do**
- 11 $b_{k,j}^i \leftarrow 0$;
- 12 **for** $v = 0, 1, \dots, 2^{16} - 1$ **do**
- 13 $b_{k,j}^i \leftarrow b_{k,j}^i + (-1)^{v_j} \cdot \mathcal{D}[v]$;
- 14 **if** $b_{k,j}^i > \iota$ **then**
- 15 Add $16k + j$ and $|b_{k,j}^i|$ to $\mathcal{S}_\iota([i])$

5 Applications to Several ARX Primitives

In this section, we apply our method to SPECK family, Alzette, SipHash, and Chaskey. Some of the results are compared in Table 1. All relevant data are provided in the supplementary material.

5.1 Application to Speck

We present improved DL distinguishers for 11/12/13-round SPECK64 and 15-round SPECK96, and propose the first 11-round DL distinguisher for SPECK32 with absolute correlation $> 2^{-16}$, the first 12- and 16-round DL distinguishers of SPECK48 and SPECK96 respectively.

DL Distinguishers of Speck64. We demonstrate our method by searching for improved 11/12/13-round DL distinguishers of SPECK64.

New 11- and 12-round DL distinguishers. We have succeeded in improving the 11- and 12-round DL distinguishers of SPECK64 under the parameters settings $(r_1, r_m, r_2) = (3, 5, 3)$ and $(r_1, r_m, r_2) = (3, 6, 3)$ respectively.

Step 1. Run Algorithm 2 with $n = 64$, $r_1 = 3$ and $r_2 = 3$ to prepare 14 tables $\mathcal{T}^{wt_0}, \mathcal{T}^{wt_1}, \dots, \mathcal{T}^{wt_{13}}$ storing $64 \times 126 \approx 2^{13} (\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$, where $wt_0 = 13$, $wt_1 = 14, \dots, wt_{12} = 25$, $wt_{13} = 26$. As a result, a total of 6 4-tuples are stored in $\mathcal{T}^{wt_0}$, 42 ones in $\mathcal{T}^{wt_1}$, ..., 10 ones in $\mathcal{T}^{wt_{12}}$, and 21 ones in $\mathcal{T}^{wt_{13}}$.

Step 2. Experiments suggest the critical-round number for DLAs of SPECK64 can take $cr = 6$. For $r_m = 5$ and 6 (both $\leq cr$), we evaluate 5- and 6-round DLAs with Algorithm 3 by using 2^{32} pseudo-ciphertext pairs and 100 keys.

This yields new 11- and 12-round DL distinguishers for SPECK64 as

$$\begin{aligned} (10041080, 100090) &\xrightarrow[2^{-7}]{3DC} (v_1[29]) \xrightarrow[2^{-1.82}]{5DL} (v_1[13]) \xrightarrow[2^{-3}]{3LC} (400841, 400048) \\ (10041080, 100090) &\xrightarrow[2^{-7}]{3DC} (v_1[29]) \xrightarrow[2^{-9.29}]{6DL} (v_1[5]) \xrightarrow[2^{-3}]{3LC} (4580400c, 48004000) \end{aligned}$$

with total absolute correlations estimated as $|bp\varepsilon^2| = 2^{-14.82}$ and $2^{-22.29}$ respectively, improving over the previous best results in [32].

Complexity Analysis: For the above concrete example of 11-round SPECK64, we pre-filter $6 + 42 = 48$ candidates from 64×126 hourglass(-like) difference-mask pairs, reducing the direct search complexity from $\mathcal{O}(64 \times 126 \times 2^{32} \times 100) \approx \mathcal{O}(2^{52})$ to $\mathcal{O}(48 \times 2^{32} \times 100) \approx \mathcal{O}(2^{44})$.

New 13-round DL distinguisher. Following Remark 3 for choosing the decomposition parameters, we construct 13-round hourglass(-like) structural DL distinguishers with $(r_1, r_m, r_2) = (3, 7, 3)$. For $r_m = 7 (> cr)$, we perform the component-wise search via Algorithm 4 using 14 tables $\mathcal{T}^{wt_0}, \mathcal{T}^{wt_1}, \dots, \mathcal{T}^{wt_{13}}$ constructed above.

Step 1. Set an absolute DL correlation threshold $\iota = 2^{-8}$. Run Algorithm 6 or Algorithm 5 with parameters $n = 64$, $\iota = 2^{-8}$, $r_m = 7$, and $t = 1$ to obtain SUBs sets $\mathcal{S}_\iota(\delta)$ of 6-round DLAs for all 1-bit differences $\delta = [i], i = 0, \dots, 63$.

Step 2. With threshold $\chi = 2^{-1}$, we execute Algorithm 4 to identify high-correlation 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$. The *for* loop in lines 7-10 yield no candidates from $\mathcal{T}^{wt_i}$ for $wt_i = 13, 14, 15, 16$, and three for $wt_i = 17$:

$$(v_0[23]) \xrightarrow{7DL} (v_1[2]), (v_0[15]) \xrightarrow{7DL} (v_1[5]), (v_0[31]) \xrightarrow{7DL} (v_1[10, 9]).$$

Our experiments over 2^{34} pseudo-ciphertext pairs and 100 keys show the 7-round DLA $(v_0[15]) \rightarrow (v_1[5])$ yields the highest absolute correlation $2^{-9.59}$, thus we update $\tau = 26.59 (17 + 9.59)$. Subsequent searches find no candidates in $\mathcal{T}^{wt_i}$ for $wt_i = 18$, one candidate for $wt_i = 19$: $(v_0[15]) \xrightarrow{7DL} (v_1[26])$, and four candidates for $wt_i = 20$: $(v_0[10]) \xrightarrow{7DL} (v_1[0])$, $(v_0[7]) \xrightarrow{7DL} (v_1[7])$, $(v_0[7]) \xrightarrow{7DL} (v_1[18])$, and $(v_0[7]) \xrightarrow{7DL} (v_1[29])$. We iteratively evaluate these candidates and constantly updating τ until the loop terminates.

Following the above steps, we construct a new 13-round DL distinguisher

$$\delta'_{opt} \xrightarrow[2^{-11}]{3DC} (v_0[15]) \xrightarrow[2^{-9.59}]{7DL} (v_1[5]) \xrightarrow[2^{-3}]{3LC} \lambda'_{opt},$$

with absolute correlation estimated as $|bp\epsilon^2| = 2^{-26.59}$, improving over the previous result in [14] with absolute correlation $2^{-28.15}$.

DL Distinguishers of Speck96. For SPECK96, we improve the 15-round DL distinguisher and propose the first 16-round one (details in Supp. Mat. I).

New 15-round DL distinguisher. The improved 15-round DL distinguisher is constructed under $(r_1, r_m, r_2) = (4, 7, 4)$ as

$$\delta'_{opt} \xrightarrow[2^{-17}]{4DC} (v_1[43]) \xrightarrow[-2^{-9.36}]{7DL} (v_1[10, 9]) \xrightarrow[2^{-6}]{4LC} \lambda'_{opt},$$

where the middle 7-round DLA is experimentally evaluated using 2^{35} pseudo-ciphertext pairs and 100 keys. Thus the total correlation is $|bp\epsilon^2| = 2^{-38.36}$, improving over the previous best result $2^{-41.72}$ from [14].

The first 16-round DL distinguisher. We construct a 16-round DL distinguisher under $(r_1, r_m, r_2) = (4, 8, 4)$ as

$$\delta'_{opt} \xrightarrow[2^{-22}]{4DC} (v_0[7]) \xrightarrow[-2^{-10.32}]{8DL} (v_1[26]) \xrightarrow[2^{-8}]{4LC} \lambda'_{opt},$$

where the middle 8-round DLA is evaluated using 2^{36} pseudo-ciphertext pairs and 50 keys. To obtain a tighter bound, the 4-round LC correlation with output mask $\lambda'_{opt} = (404c41cd20, 4040404100)$ is verified as $2^{-7.73}$, thus we derive a better estimations for the total correlation as $2^{-47.58}$, which is the first DL distinguisher for 16-round SPECK96.

DL Distinguishers of Speck32 and Speck48. We propose the first (valid) 11- and 12-round DL distinguisher of SPECK32 and SPECK48 respectively.

Speck32. As stated in Section 3.3, the previous best 11-round DL distinguisher of SPECK32 given in [9] can be re-decomposed into the $(3, 4, 4)$ -hourglass structural DL distinguisher as shown in Eq. (2), with absolute correlation estimated as $|bp\epsilon^2| = 2^{-16.19}$, which is strictly not a valid distinguisher as $1/(bp\epsilon^2)^2 > 2^{32}$. As shown in Eq. (3) and Eq. (4), we construct two new DL distinguishers with absolute correlations $|bp\epsilon^2| = 2^{-15.89}$ and $2^{-15.83}$ (both strictly higher than 2^{-16}) respectively, which are the first valid DL distinguishers for 11-round SPECK32.

Speck48. With our new search method, we construct two $(3, 6, 3)$ -hourglass(-like) structural DL distinguishers for 12-round SPECK48

$$\begin{aligned} & (d240, 420050) \xrightarrow[2^{-11}]{3DC} (v_0[15]) \xrightarrow[-2^{-6.40}]{6DL} (v_1[2]) \xrightarrow[2^{-3}]{3LC} (8a801, 90800), \\ & (400052, 504200) \xrightarrow[2^{-11}]{3DC} (v_0[7]) \xrightarrow[-2^{-6.68}]{6DL} (v_1[5]) \xrightarrow[2^{-3}]{3LC} (41400c, 4c4000), \end{aligned}$$

with absolute correlations $|bp\epsilon^2| = 2^{-23.40}$ and $2^{-23.68}$ respectively.

DL Distinguishers of Speck128. We have verified that the 18-round DL distinguisher of SPECK128 presented in [14] can also be constructed using our new search method. Besides, though we do not find better DL distinguishers, we construct a new one

$$\delta'_{opt} \xrightarrow[2^{-31}]{5DC} (v_0[50]) \xrightarrow[-2^{-10.85}]{9DL} (v_1[2]) \xrightarrow[2^{-7}]{4LC} \lambda'_{opt},$$

with total correlation $|bp\epsilon^2| = 2^{-55.85}$, slightly worse than that constructed in [14] with absolute correlation $2^{-55.81}$.

5.2 Application to Alzette

We improve all DL distinguishers from [32,37] for different reduced-round Alzette. In particular, we construct the first 11- to 14-round DL distinguishers, up to 4 rounds longer than the previous best. We refer to Supp. Mat. J for detailed search process, along with the comparisons as shown in Table 10 and Table 11.

11-round. We construct a (2, 6, 3)-hourglass(-like) structural DL distinguisher for 11-round Alzette with absolute correlation $|bp\varepsilon^2| = 2^{-14.53}$, shown as

$$(80020100, 10080) \xrightarrow[2^{-2}]{2\text{DC}} (v_1[16]) \xrightarrow[-2^{-6.53}]{6\text{DL}} (v_1[6, 5]) \xrightarrow[2^{-3}]{3\text{LC}} (\text{c0a04031}, 40\text{c08061}).$$

12-round. We have tried different combinations of decomposition parameters satisfying $r_1 + r_m + r_2 = 12$. Using Algorithm 4 with parameter $(t, \iota, \chi) = (1, 2^{-9}, 2^{-1})$, we construct a (2, 7, 3)-hourglass(-like) structural DL distinguisher for 12-round Alzette with absolute correlation $|bp\varepsilon^2| = 2^{-19.02}$, shown as

$$(04040000, 06000000) \xrightarrow[2^{-4}]{2\text{DC}} (v_1[25]) \xrightarrow[2^{-11.02}]{7\text{DL}} (v_1[15, 14]) \xrightarrow[2^{-2}]{3\text{LC}} (\text{c041c101}, 8100\text{c041}).$$

13- and 14-round. Similarly, we construct a 13-round DL distinguisher as

$$(\text{a8400050}, 40201028) \xrightarrow[2^{-11}]{3\text{DC}} (v_0[29]) \xrightarrow[2^{-9.32}]{7\text{DL}} (v_1[1]) \xrightarrow[2^{-2}]{3\text{LC}} (\text{c2820001}, 810100\text{c2}),$$

with absolute correlation $|bp\varepsilon^2| = 2^{-24.32}$. Based on the same DLA, we also construct a (3, 7, 4)-hourglass(-like) structural DL distinguisher for 14-round Alzette under the 4-round LC $(v_1[1]) \xrightarrow[2^{-6}]{4\text{LC}} (8\text{b0a0205}, 81020484)$ (round 10-14). For tighter bound, $|\varepsilon|$ is experimentally confirmed as $2^{-5.47}$, thus we derive a better estimation of the total correlation as $2^{-31.26}$.

5.3 Application to SipHash

We focus on the finalization process of SipHash-2-4 [1], where four rounds are applied and the output branches are XORed together. In [37], 3- and 4-round DL distinguishers are obtained with experimental correlations $2^{-0.78}$ and $2^{-6.03}$ respectively. Using our new search method, we have constructed significantly improved 3- and 4-round ones, including deterministic ones with correlation ± 1 . We refer to Supp. Mat. K for the detailed process of the construction.

Deterministic 3-round DL distinguishers. We construct many *deterministic* (correlation ± 1) 3-round DL distinguishers under $(r_1, r_m, r_2) = (0, 2.5, 0.5)$, and one among them is $(v_2[63]) \xrightarrow[2^0]{2.5\text{DL}} (v_1[1]) \xrightarrow[2^0]{0.5\text{LC}} (v_1[18], v_2[50])$, with correlation estimated as 1.

New 4-round DL distinguishers. By choosing $r_1 = 0$, $r_m = 3$, and $r_2 = 1$, we construct our best 4-round DL distinguisher as

$$(v_2[6]) \xrightarrow[2^{-0.13}]{3\text{DL}} (v_1[19]) \xrightarrow[2^0]{1\text{LC}} (v_0[21, 0], v_1[49], v_2[17], v_3[21]),$$

whose experimental correlation is 0.911, i.e., $2^{-0.13}$, significantly improving the previous best one in [37] with absolute correlation $2^{-6.03}$.

5.4 Application to Chaskey

In [7,30], two (1.5, 4, 1.5)-hourglass(-like) structural DL distinguishers are constructed for 7-round Chaskey with absolute correlations both estimated as $|bp\varepsilon^2| = 2^{-36.11}$. Using our new method, we construct an improved 7-round one with absolute correlation $|bp\varepsilon^2| = 2^{-33.65}$, shown as

$$\delta'_{opt} \xrightarrow[2^{-17}]{1.5\text{DC}} (v_0[31]) \xrightarrow[-2^{-8.65}]{4\text{DL}} (v_3[11]) \xrightarrow[2^{-4}]{1.5\text{LC}} \lambda'_{opt}.$$

6 New DL Distinguishers against Latin Dances Ciphers

In this section, we apply our method to ARX-based Latin dance stream ciphers ChaCha and Salsa, along with a new Forró cipher [15]. Results are compared in Table 2. All related supporting data are provided in the supplementary materials.

6.1 Application to ChaCha

We improve 7- and 7.25-round DL distinguishers for ChaCha under the most commonly adopted “3.5DL+3.5LC” pattern, surpassing previous results from [8,15,17]. For a fair comparison, we compare only DL distinguishers also constructed based on the experimentally variable 3.5-round DLAs. Previous results on ChaCha are summarized in Supp. Mat. L.

Each state of ChaCha has 16 branches, denoted as v_k , $k = 0, \dots, 15$. For DL cryptanalysis, we cannot control the branches v_k for $0 \leq k \leq 11$ of the initial 4×4 state matrix as fixed secret key and constants are loaded into them. However, branches v_{12} , v_{13} , v_{14} and v_{15} can be varied as counter or nonces are loaded into them. It implies that active bit positions for input differences are limited to these 4 branches. Due to the good diffusion property of the round function of ChaCha, it's not possible that a 1-bit output difference occurs after E_1 .

Improving 7-round DL Distinguishers. We primarily focus on enhancing the 7-round DL distinguishers by leveraging experimentally verifiable 3.5-round DLAs with 1-bit input differences and output masks as done in most of the previous work [7,15,16,17,19]. While constructing DL distinguishers based on 4-round DLAs could theoretically yield stronger results, we refrain from pursuing this direction due to computational limitations. Specifically, directly experimentally evaluating 4-round DLAs is infeasible with our current resources because their extremely small absolute correlations require either impractical computational power or reliance on assumptions, as seen in prior work [8]. Nevertheless, our analysis demonstrates that the same “3.5DL+3.5LC” pattern starting from 1-bit input differences can still produce better DL distinguishers than 2-bit input differences, thereby challenging the claim made in [8].

According to the symmetry in the round update for the state of ChaCha, we know that the r_m -round DLA $v_{12}[i] \rightarrow v_q[0]$ is equivalent to the following ones:

$$v_{13}[i] \rightarrow v_{(q+1) \bmod 4}[0], \quad v_{14}[i] \rightarrow v_{(q+2) \bmod 4}[0], \quad v_{15}[i] \rightarrow v_{(q+3) \bmod 4}[0]$$

In the following, we take $\delta = (v_{13}[i])$, $i = 0, \dots, 31$ and $\lambda = (v_q[0])$, $q = 0, \dots, 15$, as the input differences and the output masks of E_m respectively.

First, we use the MILP-based automatic tool to search the optimal 3.5-round LCs with input mask $\lambda = v_q[0]$, starting at round 3.5 and ending at round 7. For this, we have the following results: The optimal 3.5-round LCs with input mask $\lambda = v_q[0]$ have the absolute correlation 2^{-37} for $q \in \{8, 9, 10, 11\}$, 2^{-40} for $q \in \{4, 5, 6, 7\}$, and 2^{-48} for $q \in \{0, 1, 2, 3, 12, 13, 14, 15\}$.

Second, we prepare the SUBs sets for all 1-bit input differences $\delta = (v_{13}[i])$, $i = 0, \dots, 31$. As stated in [7], the optimal 1-round DC is of the form

$v_{13}[i] \xrightarrow{1\text{DC}} (v_1[i+28], v_5[i+31, i+23, i+11, i+3], v_9[i+24, i+16, i+4], v_{13}[i+24, i+4])$, whose differential probability is 2^{-3} for $i = 15$, 2^{-4} for $i = 3, 7, 27$ and 2^{-5} otherwise. Let $\hat{\delta}^{(i)} \triangleq (v_1[i+28], v_5[i+31, i+23, i+11, i+3], v_9[i+24, i+16, i+4], v_{13}[i+24, i+4])$. Instead of preparing SUBs set for 3-round DLAs for each $v_{13}[i]$, we set $\iota = 2^{-12}$ to identify SUBs set $S_\iota(\hat{\delta}^{(i)})$ of each $\hat{\delta}^{(i)}$ for 2-round DLAs from round 1 to round 3 by using Algorithm 5 (with the block size $n = 512$), where 2^{30} pseudo-ciphertext pairs are used for experimental evaluations.

Finally, we execute Algorithm 4 with parameters $t = 0.5$ and $\chi = 2^{-1}$ to identify difference-mask pair candidates for experimental evaluations. For instance, we find $\hat{\delta}^{(26)}$ (round 1) and $\lambda = (v_9[0])$ (round 3.5) collide with the multi-bit mask $\hat{\lambda} = (v_3[16, 15], v_4[16], v_9[0], v_{14}[16])$ (round 3), yielding the candidate 2.5-round DLA $\hat{\delta}^{(26)} \xrightarrow{2.5\text{DL}} (v_9[0])$. Additional experimentally verified difference-mask pair candidates include $(\hat{\delta}^{(6)}, (v_{11}[0]))$, $(\hat{\delta}^{(14)}, (v_{11}[0]))$, and $(\hat{\delta}^{(10)}, (v_8[0]))$. Consequently, we construct new 7-round DL distinguishers of the form

$$\underbrace{(v_{13}[26]) \xrightarrow[2^{-5}]{1\text{DC}} \hat{\delta}^{(26)} \xrightarrow[2^{-18.06}]{2.5\text{DL}} (v_9[0])}_{3.5\text{DL}} \xrightarrow[2^{-37}]{3.5\text{LC}} \mathcal{B}, \quad (5)$$

where the 2.5-round DL correlation is experimentally verified using 2^{43} samples as $2^{-18.06}$, and thus the total absolute correlation can be estimated as $|bp\varepsilon^2| = 2^{-97.06}$. There exists many optimal 3.5-round LCs with the same input mask $(v_9[0])$ having absolute theoretical correlation 2^{-37} . According to the symmetry in the round update for the state matrix of ChaCha, we identify an output mask $\mathcal{B}$ (related to $\mathcal{A}$ of Eq. (9)) yielding the absolute experimental correlation $2^{-34.6}$. This gives us a 7-round DL distinguisher with complexity $2^{(18.06+5+34.6 \times 2) \times 2} = 2^{184.52}$, improving over the previous one under the same “3.5DL+3.5LC” pattern with the complexity $2^{189.9}$ in [8], which used a 2-bit input difference at the beginning. See Supp. Mat. M for masks $\mathcal{B}$ and the later mentioned $\mathcal{D}$, $\mathcal{E}'$, $\mathcal{G}$.

Improving 7.25-round DL Distinguishers. The 0.25-round update of the ChaCha permutation is given in Eq.(8) of Supp. Mat. A.7. Based on the 3.5-round DLAs, Dey and Sarkar [17] proposed a DL distinguisher for 7.25-round ChaCha under the “3.5DL+3.75LC” pattern, shown as

$$\underbrace{(v_{14}[6]) \xrightarrow[2^{-5}]{1\text{DC}} (v_2[2], v_6[29, 17, 9, 5], v_{10}[30, 22, 10], v_{14}[30, 10])}_{3.5\text{DL}} \xrightarrow[2^{-28.65}]{2.5\text{DL}} (v_8[0]) \xrightarrow[2^{-58}]{3.75\text{LC}} \mathcal{C}.$$

Thus, the total correlation is estimated as $|bp\varepsilon^2| = 2^{-149.65}$. In [17], the optimal 3.75-round LC with input mask $(v_8[0])$ and the given output mask $\mathcal{C}$ was experimentally evaluated as $2^{-48.21}$. Note that such a small LC correlation was achieved by dividing the 3.75-round LC (round 3.5-7.25) into 8 sub-trails (round 3.5-4, round 4-5, round 5-6, round 6-6.25, round 6.25-6.5, round 6.5-6.75, round 6.75-7, round 7-7.25) and verifying each independently. Applying the Piling-up Lemma yields an estimated 3.5-round LC correlation of $2^{-48.21}$, increasing the absolute correlation estimate to $2^{-130.07}$ (still below the 2^{-128} threshold). We note that [17] overestimated this value by a factor of 2^{12} (see [17] for details).

We construct new 7.25-round DL distinguishers using our newly found 3.5-round DLA $(v_{13}[26]) \xrightarrow{3.5\text{DL}} (v_9[0])$. The optimal 3.75-round LCs with input mask $(v_9[0])$, found via MILP, have the absolute theoretical correlation 2^{-48} . Thus we construct new 7.25-round DL distinguishers of the form

$$(v_{13}[26]) \xrightarrow[2^{-5}]{1\text{DC}} \underbrace{\hat{\delta}^{(26)} \xrightarrow[2^{-18.06}]{2.5\text{DL}} (v_9[0])}_{3.5\text{DL}} \xrightarrow[2^{-48}]{3.75\text{LC}} \mathcal{D}, \quad (6)$$

with total correlation estimated as $|bp\varepsilon^2| = 2^{-119.06}$. Our experimental verifications show that one of the optimal 3.75-round LCs with a given output mask $\mathcal{D}$ has the absolute correlation $2^{-47.46}$, achieved by splitting the verifications into 5 sub-trails (round 3.5-6, round 6-6.5, round 6.5-6.75, round 6.75-7, round 7-7.25) according to the theoretical LC correlation of each sub-trail⁴. Thus, we construct a 7.25-round DL distinguisher based on the 3.5-round DLA with time and data complexity $2^{(18.06+5+47.46 \times 2) \times 2} = 2^{235.96}$.

Theoretical/Mathematical Interpretation. In [18], Dey *et al.* theoretically analyzed the DL distinguisher of 3.5-round ChaCha originally found experimentally in [16], and found some mismatch between the theoretical and experimental results. According to the fact that the absolute correlation values decrease as the number of rounds increases, the theoretical demonstration of the correlation involved expressing the bit at round 3.5 in terms of those at round 3.

Inspired by the method in [18], we try to provide a mathematical explanation in support of the experimental correlation value $2^{-18.06}$ of the new 2.5-round DLA in Eq.(6). Generally, we express the bit at round 3.5 in terms of a bit at round 3 and another bit at round 3.5. Let $X^{(r)}$ be the state matrix after r rounds. We use x_q^r to denote the q -th word of $X^{(r)}$ and $x_q^r[i]$ the i -th bit of x_q^r . According to the update function, it holds that $x_q^{3.5}[0] = x_q^3[0] \oplus x_{q'}^{3.5}[0]$, where $(q, q') \in \{(8, 13), (9, 14), (10, 15), (11, 12)\}$. Let Δx_q^r be the difference of x_q^r and $x_q'^r$, i.e., $\Delta x_q^r = x_q^r \oplus x_q'^r$, and $\Delta x_q^r[i]$ be the i -th bit of Δx_q^r , then we have

$$\Delta x_q^{3.5}[0] = \Delta x_q^3[0] \oplus \Delta x_{q'}^{3.5}[0], \quad (q, q') \in \{(8, 13), (9, 14), (10, 15), (11, 12)\}.$$

In our case, we can derive the correlation $b \triangleq \mathcal{C}(\hat{\delta}^{(26)} \xrightarrow{2.5\text{DL}} (v_9[0]))$ following the probability approach. For short, we define $b_0 = \mathcal{C}(\hat{\delta}^{(26)} \xrightarrow{2.5\text{DL}} (v_{14}[0]))$, and let $b_{10} \triangleq \mathcal{C}(\hat{\delta}^{(26)} \xrightarrow{2\text{DL}} (v_9[0]) | \Delta x_{14}^{3.5}[0] = 0)$ and $b_{11} \triangleq \mathcal{C}(\hat{\delta}^{(26)} \xrightarrow{2\text{DL}} (v_9[0]) | \Delta x_{14}^{3.5}[0] = 1)$ represent the correlation of $\hat{\delta}^{(26)} \xrightarrow{2\text{DL}} (v_9[0])$ conditioned on $\Delta x_{14}^{3.5}[0] = 0$ and

⁴ See the auxiliary file for these sub-trails and corresponding experimental correlations.

$\Delta x_{14}^{3.5}[0] = 1$ respectively. According to the total probability formula, we have

$$\begin{aligned} \Pr(\Delta x_9^{3.5}[0]=0) &= \Pr(\Delta x_9^3[0] \oplus \Delta x_{14}^{3.5}[0]=0) \\ &= \Pr(\Delta x_{14}^{3.5}[0]=0) \cdot \Pr(\Delta x_9^3[0]=0 | \Delta x_{14}^{3.5}[0]=0) + \Pr(\Delta x_{14}^{3.5}[0]=1) \cdot \Pr(\Delta x_9^3[0]=1 | \Delta x_{14}^{3.5}[0]=1) \\ &= (1+b_0)/2 \cdot (1+b_{10})/2 + (1-b_0)/2 \cdot (1-b_{11})/2 \end{aligned}$$

Then we derive $b = \frac{1}{2}((b_{10} - b_{11}) + b_0(b_{10} + b_{11}))$. Through the experiments over 2^{41} samples, we obtain $b_0 = -0.0000344899$. Besides, we have, given $\Delta x_{14}^{3.5}[0] = 0$ and $\Delta x_{14}^{3.5}[0] = 1$, the correlations of $\delta^{(26)} \xrightarrow{2\text{DL}} (v_9[0])$ are $b_{10} = 0.000479151$ and $b_{11} = 0.000478301$ respectively. Above all, the correlation of $\hat{\delta}^{(26)} \xrightarrow{2.5\text{DL}} (v_9[0])$ is derived as $((b_{10} - b_{11}) + b_0(b_{10} - b_{11}))/2 = 2^{-21.22}$. By experiments we find this value to be approximately $2^{-18.06}$, only a little bit of difference between this mathematically obtained result from the fully experimental result. Thus we obtain a loose bound for our new 7- and 7.25-round DL distinguishers as $2^{-21.22} \times 2^{-5-37 \times 2} = 2^{-100.22}$ and $2^{-19.48} \times 2^{-5-48 \times 2} = 2^{-122.22}$ respectively.

6.2 Improving 8-round DL Distinguishers of Salsa

In [15], by applying the BLE technique, Coutinho *et al.* propose the first 8-round DL distinguisher of Salsa based on a single-bit DLA reaching 5 rounds, which can be expressed as follows:

$$\underbrace{(\mathbf{v}_7[\mathbf{31}]) \xrightarrow[2^{-1}]{1\text{DC}} (v_7[31], v_{11}[12], v_{15}[30, 17]) \xrightarrow{3\text{DL}} (v_0[0], v_4[7], v_{12}[0]) \xrightarrow[2^0]{1\text{LC}} (\mathbf{v}_4[\mathbf{7}]) \xrightarrow[2^{-34}]{3\text{LC}} \mathcal{E}}_{5\text{DL}}.$$

While it is computationally infeasible to directly experimentally estimate the 5-round DLA with difference-mask pair $(v_7[31], v_4[7])$ due to the quite small absolute LC correlation, [15] experimentally evaluated the 3-round DLAs (round 1-4) with input difference $\delta^{(1)} = (v_7[31], v_{11}[12], v_{15}[30, 17])$ and *three* output masks $(v_0[0])$, $(v_4[7])$, and $(v_{12}[0])$ respectively, and used the Piling-up Lemma to reach a DL correlation (round 1-5) as $2^{-42.01}$. Besides, they experimentally verified the 3-round LC (round 5-8) with the given output mask $\mathcal{E}$ as $2^{-32.65}$. When analyzing the distinguishing attack complexity, [15] used a technique proposed by Beierle *et al.* [7] to reduce the complexity by a factor of p^{-1} , giving a distinguisher for 8-round Salsa with time and data complexity of $2 \times 2^{(42.01+32.65 \times 2) \times 2} = 2^{215.62}$. For more details, We refer to Section 4 of [15].

Recently, Dey *et al.* [20] re-estimated the 5-round DL correlation by introducing one more condition in the second round. Specifically, they first propagated the difference $\delta^{(1)}$ to $\delta^{(1.25)}$ for 0.25-round with probability 2^{-2} , and then experimentally evaluated the 2.75-round DLAs (round 1.25-4) with input difference $\delta^{(1.25)}$ and *two* output masks $v_0[0]$ and $(v_4[7], v_{12}[0])$ respectively, and finally used the Piling-up Lemma to reach a DL correlation (round 1-5) as $2^{-2} \times 2^{-33.75} = 2^{-35.75}$. While [20] focused on improving the key recovery attacks considering the data limit, we note the data/time complexity of the distinguishing attack using the same methodology as in the compared paper [15] would be $2 \times 2^{(35.75+32.65 \times 2) \times 2} = 2^{203.1}$ without considering the data limit.

Here we improve the 8-round DL distinguisher by using our new search algorithm. For the DL part, one one hand, we tried to find new 5-round single-bit

DLAs by constructing SUBs sets from 1-th round to 3.75-th round. But unfortunately, we were not able to find better ones but find for the first time two slightly worse ones with difference-mask pairs $(v_7[0], v_4[8])$ and $(v_7[27], v_4[3])$ respectively. On the other hand, we re-estimated the 5-round DL correlation following a similar idea of [20] by considering another intermediate difference $\delta^{(1.25)}$ propagated from $\delta^{(1)}$ with the same probability 2^{-2} , reaching a DL correlation (round 1-5) as $2^{-2} \times 2^{-33.75} + 2^{-2} \times 2^{-35.18} = 2^{-35.29}$. For the LC part, we have found our best 3-round LCs with the absolute theoretical correlation 2^{-31} and one of such with output mask $\mathcal{E}'$ has absolute experimental correlation $2^{-30.13}$. These improvements give us a new 8-round DL distinguisher with time and data complexity $2 \times 2^{(35.29+30.13 \times 2) \times 2} = 2^{192.1}$, improving over those in [15] and [20] by a factor of 2^{23} and 2^{11} respectively.

6.3 New and Improved DL Distinguishers of Forró

Forró [15] is a new stream cipher proposed at ASIACRYPT 2022, having the same security as ChaCha with fewer operations. In [15], the designers derived the best distinguisher for 5.25 rounds of Forró, constructed as $(v_5[11]) \xrightarrow[-2^{-10.88}]{2DL} (v_{10}[0]) \xrightarrow[2^{-47}]{3.25LC} \mathcal{F}$, where the absolute theoretical correlation of the 3.25-round LC⁵ is 2^{-47} , while the absolute experimental correlation is verified as $2^{-39.45}$ by splitting it into sub-trails. It gives a distinguisher for 5.25-round Forró with complexity⁶ $2^{(10.88+39.45 \times 2) \times 2} = 2^{179.56}$. See Section 5 of [15] for details.

Here we improve the 5.25-round distinguisher of Forró; and more importantly, we propose the first 5.5-round ones. Using the same 2-round DLA with difference-mask pair $(v_5[11], v_{10}[0])$, we find our best 3.25-round LCs with absolute theoretical correlation 2^{-44} , and one of such has the absolute experimental correlation $2^{-38.97}$, yielding a slightly better distinguisher with data and time complexity $2^{(10.88+38.97 \times 2) \times 2} = 2^{177.64}$. Besides, we find our best 3.5-round LCs with absolute theoretical correlation 2^{-60} , and construct 5.5-round DL distinguishers in the form of $(v_5[11]) \xrightarrow[-2^{-10.88}]{2DL} (v_{10}[0]) \xrightarrow[2^{-60}]{3.5LC} \mathcal{G}$. To obtain a tighter bound, we also experimentally verify these 3.5-round LCs, and one of such with output mask $\mathcal{G}$ has absolute correlation $2^{-55.19}$. Based on this, the complexity of attack on 5.5-round Forró becomes $2^{(10.88+55.19 \times 2) \times 2} = 2^{242.52}$, improving over the result 2^{296} given in [15] by a factor of 2^{53} , which is also the first effective 5.5-round DL distinguisher for Forró.

7 Conclusion

In this paper, we identify for the first time the hourglass(-like) structural commonalities among DL distinguishers on ARX ciphers, based on our systematic re-decomposition technique and thorough investigations on DL distinguishers of SPECK32 and Alzette by a comprehensive set of experiments. Building on

⁵ Our MILP-based algorithm yields this 3.25-round LC correlation as 2^{-46} (vs. 2^{-47}).

⁶ [15] overestimates this complexity by misusing the 2-round DL correlation value b .

this insight, we develop an efficient and generalized approach to construct good hourglass(-like) structural DL distinguishers, which we apply to a wide range of ARX-based primitives, including the SPECK family, Alzette, SipHash, Chaskey, and Latin dance stream ciphers, yielding new state-of-the-art DL distinguishers or improving upon existing ones. The newly-found distinguishers have not only verified the effectiveness of our method but also indicated the power of the classic hourglass(-like) structure in conducting DL attacks on ARX primitives.

Future work. While this paper focuses on constructing longer DL distinguishers, an important direction is investigating whether hourglass(-like) structures can optimize distinguishers for key recovery attacks. We leave it as a future work due to the space limits, and we also anticipate further research on broader applications and deeper theoretical analysis on hourglass(-like) structure.

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SUPPLEMENTARY MATERIAL

A Descriptions of the Involved ARX Designs

A.1 Speck Family

SPECK [4] is a family of ARX block ciphers designed by the U.S. National Security Agency (NSA). The SPECK family has been a part of the RFID air interface standard (ISO/IEC 29167-22). Ten members are contained in SPECK family, each of which is characterized by its block size $2n$ and key size mn , thus is named as $\text{Speck}_{2n/mn}$. All the members with the same block size are also named SPECK_{2n} , i.e., SPECK32, SPECK48, SPECK64, SPECK96 and SPECK128 respectively. The round function of SPECK is shown in Fig.3. For SPECK32, two parameters are $R = 7$ and $L = 2$. For the remaining members, two parameters are $R = 8$ and $L = 3$.

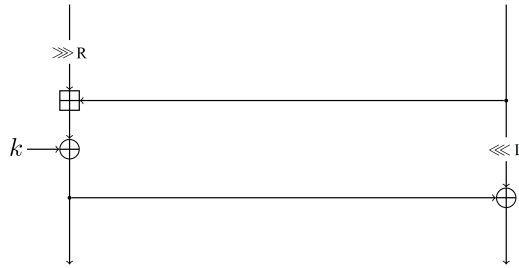


Fig. 3: The round function of Speck

A.2 Alzette

Alzette [5] is a 64-bit ARX-based S-box designed by Beierle *et al.* at CRYPTO 2020, which is the main building block of the Sparkle-suite [6], a collection of lightweight symmetric cryptographic algorithms (AEADs and hash functions) currently in the final round of the NIST lightweight cryptography standardization effort. An instance of Alzette with an input $(x, y) \in \mathbb{F}_2^{32} \times \mathbb{F}_2^{32}$ is depicted on the left-hand side of Fig.4, where the rotation constants are denoted by $(r_0, r_1, r_2, r_3) = (31, 17, 0, 24)$, $(s_0, s_1, s_2, s_3) = (24, 17, 31, 16)$. As stated in [37], the Alzette round function can also be converted into its equivalent form as shown on the right-hand side of Fig.4.

A.3 LEA

The LEA family of block ciphers was published at WISA in 2013, expecting to provide confidentiality in both high-speed and lightweight environments. In

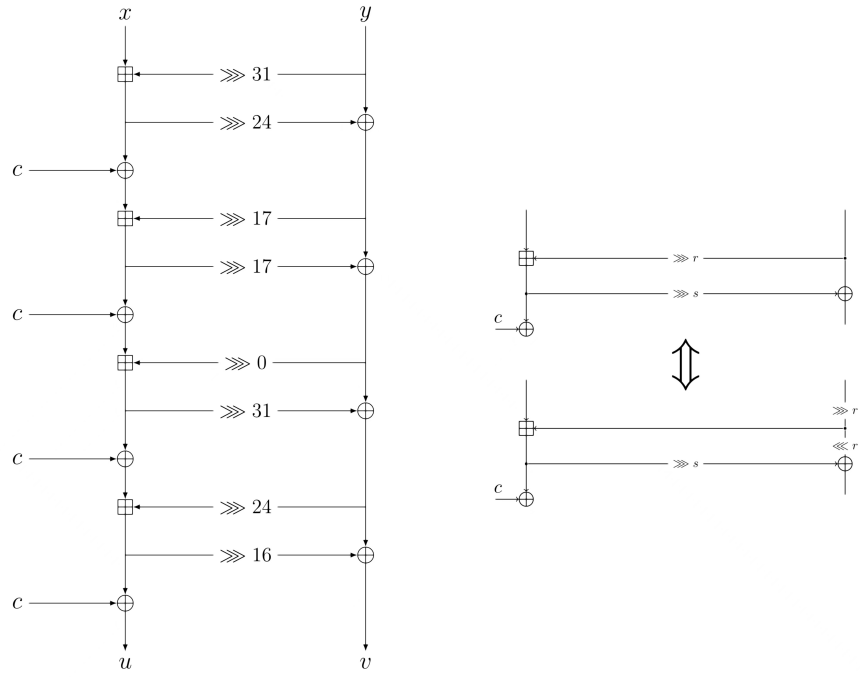


Fig. 4: On the left-hand side, the **Alzette** instance A_c . On the right-hand side, an equivalent transformation of the round function

2016, it is established as the national standard of Republic of Korea. Six years later, it is included in the ISO/IEC 29192-2:2019 standard. The LEA family has three members with a common block size of 128 bits and three different key sizes of 128, 192, and 256 bits, denoted by LEA-128, LEA-192, and LEA-256 respectively. The encryption of LEA maps a plaintext of four 32-bit words into a ciphertext using a sequence of r rounds, where $r = 24$ for LEA-128, $r = 28$ for LEA-192 and $r = 32$ for LEA-256. A schematic view of the round function of LEA is shown in Fig.5.

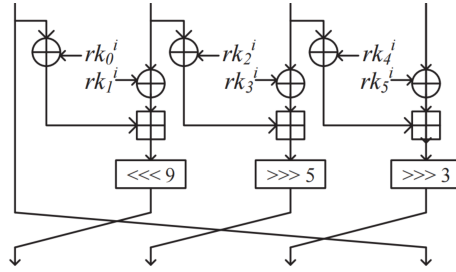


Fig. 5: The round function of LEA

A.4 Cham

CHAM [28] is a family of lightweight block ciphers designed for resource-constrained environments, such as Internet of Things (IoT) devices, embedded systems, and other applications where both low power consumption and high performance are crucial. It is a family of block ciphers with four-branch generalized Feistel structure. Each cipher in this family is denoted as CHAM- n/k , where n represents the block size and k represents the key size, such as CHAM-64/128, CHAM-128/128 and CHAM-128/256. By applying Nr iterations of the key-dependent round function, CHAM- n/k encrypts an n -bit plaintext of four $w = n/4$ -bit words to an n -bit ciphertext of four $w = n/4$ -bit words. For a round key rk_i , the i -th round function of CHAM is as shown in fig.6.

A.5 SipHash

SipHash [1] is a family of ARX-based pseudo-random functions optimized for short inputs, whose instances are widely deployed in real-world applications. The round function of SipHash is illustrated in Fig.7.

A.6 Chaskey

Chaskey [35] is a lightweight MAC algorithm with a round function depicted in Fig.8, which operates on four 32-bit words and employs 12 rounds. It is an ISO

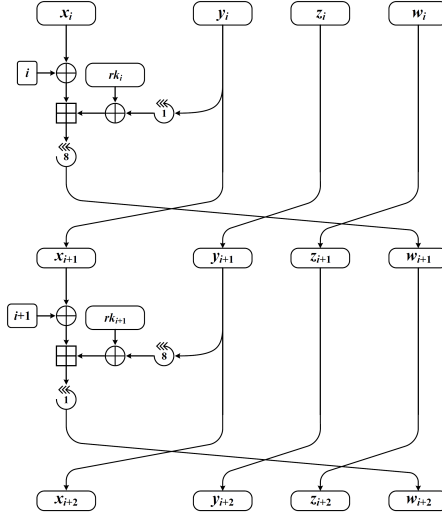


Fig. 6: The round function of CHAM

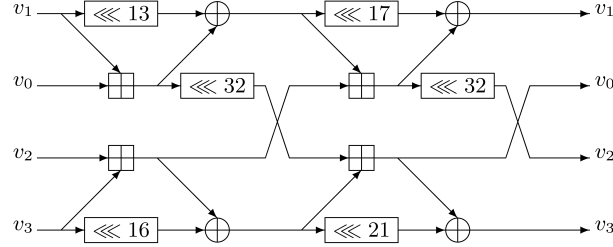


Fig. 7: The round function of SipHash

standard and is deployed in commercial products by some automotive suppliers and major industrial control systems.

A.7 ChaCha, Salsa and Forró

Salsa and ChaCha, following the names of Latin dances, are two well-known ARX-based stream ciphers designed by Bernstein, which manipulate 128-bit or 256-bit keys. Salsa [11] was designed in 2005 as a candidate for the eSTREAM competition, and Salsa20/12 was accepted into the final eSTREAM software portfolio. Salsa is especially important and is used in practice in several applications, such as DNS implementations, in the Linux Kernel, Password managers (e.g., KeePassX and MacPass), messaging software (e.g., Viber and Discord). ChaCha [10] was introduced in 2008 in line with Salsa with a modified

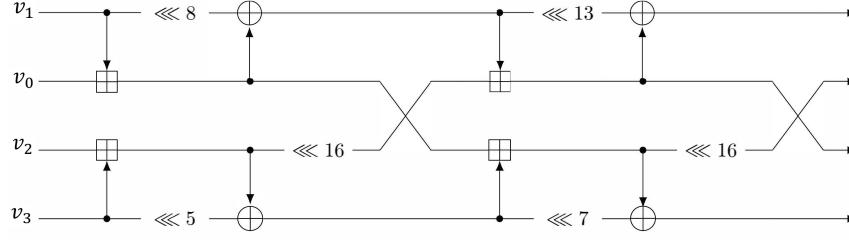


Fig. 8: The round function of Chaskey

round function, which has been widely used in many algorithms and protocols by different organizations. As easily seen from the public domain information, ChaCha20-Poly1305 is used in protocols like IPsec, SSH, TLS 1.2, DTLS 1.2, TLS 1.3, WireGuard, S/MIME 4.0, OTRv4, etc.

Salsa and ChaCha operate on a 4×4 state matrix of sixteen 32-bit words. The 256-bit key versions of the ciphers take 8 key words ($k_0, k_1, \dots, k_7$), 4 constant words $c_0 = 0x61707865$, $c_1 = 0x3320646e$, $c_2 = 0x79622d32$, $c_3 = 0xb206574$, and a total of 4 IVs consisting of nonces and counter words. The initial state matrix $X^{(0)}$ of Salsa and ChaCha and the state matrix after r rounds (denoted by $X^{(r)}$) are given respectively as follows

$$\text{Salsa : } \begin{pmatrix} c_0 & k_0 & k_1 & k_2 \\ k_3 & c_1 & v_0 & v_1 \\ v_2 & v_3 & c_2 & k_4 \\ k_5 & k_6 & k_7 & c_3 \end{pmatrix}, \text{ Chacha : } \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ k_0 & k_1 & k_2 & k_3 \\ k_4 & k_5 & k_6 & k_7 \\ t_0 & v_0 & v_1 & v_2 \end{pmatrix}, X^{(r)} = \begin{pmatrix} x_0^r & x_1^r & x_2^r & x_3^r \\ x_4^r & x_5^r & x_6^r & x_7^r \\ x_8^r & x_9^r & x_{10}^r & x_{11}^r \\ x_{12}^r & x_{13}^r & x_{14}^r & x_{15}^r \end{pmatrix}.$$

Salsa. At each round, the quarter-round (QR) function is operated on a 4-tuple $(x_a^{m-1}, x_b^{m-1}, x_c^{m-1}, x_d^{m-1})$ in the following manner, as depicted in Fig. 9.

$$\begin{aligned} x_b^{m-0.5} &= x_d^{m-1}, \quad x_c^{m-0.5} = x_a^{m-1}, \\ x_d^{m-0.5} &= ((x_a^{m-1} \boxplus x_d^{m-1}) \lll 7) \oplus x_b^{m-1}, \quad x_a^{m-0.5} = ((x_a^{m-1} \boxplus x_d^{m-0.5}) \lll 9) \oplus x_c^{m-1}, \\ x_b^m &= x_d^{m-0.5}, \quad x_c^m = x_a^{m-0.5}, \\ x_d^m &= ((x_a^{m-0.5} \boxplus x_d^{m-0.5}) \lll 13) \oplus x_b^{m-0.5}, \quad x_a^m = ((x_a^{m-0.5} \boxplus x_d^m) \lll 18) \oplus x_c^{m-0.5}. \end{aligned}$$

We call the update of $(x_a^{m-1}, x_b^{m-1}, x_c^{m-1}, x_d^{m-1})$ to $(x_a^{m-0.5}, x_b^{m-0.5}, x_c^{m-0.5}, x_d^{m-0.5})$ of the state matrix the 0.5-round update of the cipher. Based on this, four parallel applications of QR are performed on four 4-tuples formed by the words of the state matrix. Specifically, at odd-numbered rounds, the QR equations are applied to the columns of the state, i.e., $(x_0^r, x_4^r, x_8^r, x_{12}^r)$, $(x_5^r, x_9^r, x_{13}^r, x_{17}^r)$, $(x_{10}^r, x_{14}^r, x_{18}^r, x_{22}^r)$, $(x_{15}^r, x_{19}^r, x_{23}^r, x_{27}^r)$, while at even-numbered rounds, the QR equations are applied to the rows of the state instead, i.e., $(x_0^r, x_1^r, x_2^r, x_3^r)$, $(x_5^r, x_6^r, x_7^r, x_4^r)$, $(x_{10}^r, x_{11}^r, x_8^r, x_9^r)$, $(x_{15}^r, x_{12}^r, x_{13}^r, x_{14}^r)$. The keystream block Z is obtained by the additions of matrices $X^{(0)}$ and $X^{(r)}$ as $Z = X^{(0)} \boxplus X^{(r)}$.

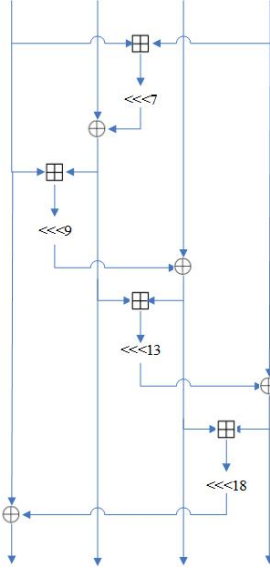


Fig. 9: The QR function of Salsa

Note that in our analysis, we are also concerned about the 0.25-round update of the state. For Salsa, the 0.25-round update from input $(x_a^m, x_b^m, x_c^m, x_d^m)$ to output $(x_a^{m+0.25}, x_b^{m+0.25}, x_c^{m+0.25}, x_d^{m+0.25})$ is given as

$$\begin{aligned} x_a^{m+0.25} &= ((x_a^m \boxplus x_d^m) \lll cst) \oplus x_b^m, \quad cst \in \{7, 9, 13, 18\} \\ x_b^{m+0.25} &= x_c^m, \quad x_c^{m+0.25} = x_d^m, \quad x_d^{m+0.25} = x_a^m. \end{aligned} \quad (7)$$

ChaCha. At each round, the QR function is operated on a 4-tuple $(x_a^{m-1}, x_b^{m-1}, x_c^{m-1}, x_d^{m-1})$ in the following manner, also as depicted in Fig. 10.

$$\begin{aligned} x_a^{m-0.5} &= x_a^{m-1} \boxplus x_b^{m-1}, \quad x_d^{m-0.5} = ((x_d^{m-1} \oplus x_a^{m-0.5}) \lll 16), \\ x_c^{m-0.5} &= x_c^{m-1} \boxplus x_d^{m-0.5}, \quad x_b^{m-0.5} = ((x_b^{m-1} \oplus x_c^{m-0.5}) \lll 12), \\ x_a^m &= x_a^{m-0.5} \boxplus x_b^{m-0.5}, \quad x_d^m = ((x_d^{m-0.5} \oplus x_a^m) \lll 8), \\ x_c^m &= x_c^{m-0.5} \boxplus x_d^m, \quad x_b^m = ((x_b^{m-0.5} \oplus x_c^m) \lll 7). \end{aligned}$$

We also call the update of $(x_a^{m-1}, x_b^{m-1}, x_c^{m-1}, x_d^{m-1})$ to $(x_a^{m-0.5}, x_b^{m-0.5}, x_c^{m-0.5}, x_d^{m-0.5})$ of the state matrix the 0.5-round update of the cipher. Similarly, four parallel applications of QR are performed on four 4-tuples formed by the words of the state matrix. At odd-numbered rounds, the QR equations are applied to the columns of the state, i.e., $(x_0^r, x_4^r, x_8^r, x_{12}^r)$, $(x_1^r, x_5^r, x_9^r, x_{13}^r)$, $(x_2^r, x_6^r, x_{10}^r, x_{14}^r)$, $(x_3^r, x_7^r, x_{11}^r, x_{15}^r)$, while at even-numbered rounds, they are applied to the diagonals of the state instead, i.e., $(x_0^r, x_5^r, x_{10}^r, x_{15}^r)$, $(x_1^r, x_6^r, x_{11}^r, x_{12}^r)$, $(x_2^r, x_7^r, x_8^r, x_{13}^r)$, $(x_3^r, x_4^r, x_9^r, x_{14}^r)$. Similarly, the keystream block Z is obtained by $Z = X^{(0)} \boxplus X^{(r)}$.

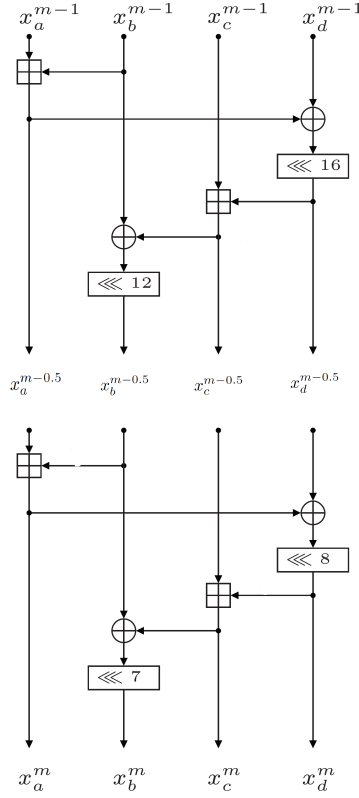


Fig. 10: The QR function of ChaCha

For ChaCha, the 0.25-round update from input $(x_a^m, x_b^m, x_c^m, x_d^m)$ to output $(x_a^{m+0.25}, x_b^{m+0.25}, x_c^{m+0.25}, x_d^{m+0.25})$ is given as

$$\begin{aligned} x_a^{m+0.25} &= x_a^m \boxplus x_b^m, \quad x_b^{m+0.25} = x_b^m, \quad x_c^{m+0.25} = x_c^m, \\ x_d^{m+0.25} &= ((x_d^m \oplus x_a^{m+0.25}) \lll cst), \quad cst \in \{16, 12, 8, 7\} \end{aligned} \quad (8)$$

Forró [15] is a new ARX-based stream cipher proposed by Coutinho *et al.* at ASIACRYPT 2022, with some modifications from ChaCha in order to provide better diffusion per round and higher resistance to cryptanalysis with fewer operations. In the design of Forró, a new technique called *Pollination* is introduced and used in the round function. At each round, the sub-round (SR) function $X^{[m]} = SR(a, b, c, d, e, X^{[m-1]})$ is defined in the following manner, also as de-

picted in Fig. 11.

$$\begin{aligned}
x_d'^{m-1} &= x_d^{m-1} \boxplus x_e^{m-1}, \quad x_c'^{m-1} = x_c^{m-1} \oplus x_d'^{m-1}, \\
x_b'^{m-1} &= (x_b^{m-1} \boxplus x_c'^{m-1}) \lll 10, \quad x_a'^{m-1} = x_a^{m-1} \boxplus x_b'^{m-1}. \\
x_e^m &= x_e^{m-1} \oplus x_a'^{m-1}, \quad x_d^m = (x_d'^{m-1} \boxplus x_e^m) \lll 27, \\
x_c^m &= x_c'^{m-1} \boxplus x_d^m, \quad x_b^m = x_b'^{m-1} \oplus x_c^m, \quad x_a^m = (x_a'^{m-1} \boxplus x_b^m) \lll 8.
\end{aligned}$$

Each round of Forró is defined in terms of its sub-rounds. More precisely, each

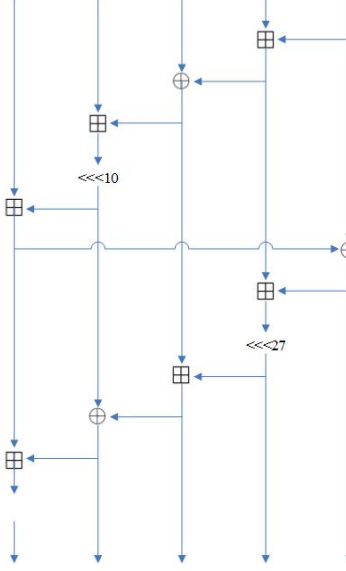


Fig. 11: The QR function of Forró

round has 4 sub-rounds and $X^{(r)} = X^{[4r]}$. Then, in an odd round, when $r \in \{1, 3, 5, 7, \dots\}$, $X^{(r)}$ is defined from $X^{(r-1)}$ in the following manner

$$\begin{aligned}
X^{[4r-3]} &= \text{SR}(0, 4, 8, 12, 3, X^{[4r-4]}), \quad X^{[4r-2]} = \text{SR}(1, 5, 9, 13, 0, X^{[4r-3]}), \\
X^{[4r-1]} &= \text{SR}(2, 6, 10, 14, 1, X^{[4r-2]}), \quad X^{[4r]} = \text{SR}(3, 7, 11, 15, 2, X^{[4r-1]}).
\end{aligned}$$

and for even-numbered rounds $r \in \{2, 4, 6, 8, \dots\}$ from

$$\begin{aligned}
X^{[4r-3]} &= \text{SR}(0, 5, 10, 15, 3, X^{[4r-4]}), \quad X^{[4r-2]} = \text{SR}(1, 6, 11, 12, 0, X^{[4r-3]}), \\
X^{[4r-1]} &= \text{SR}(2, 7, 8, 13, 1, X^{[4r-2]}), \quad X^{[4r]} = \text{SR}(3, 4, 9, 14, 2, X^{[4r-1]}).
\end{aligned}$$

B Existing DL Distinguishers for Reduced-round Speck

In [9,32,37,41], DL distinguishers for reduced-round SPECK32, SPECK48 and SPECK64 are presented. In particular, [14] recently improved the 11-round DL

distinguisher for SPECK48, and found the first DL distinguishers for 13-round SPECK64, 15-round SPECK96 and 18-round SPECK128. We have summarized all these existing DL distinguishers, as shown in Table 4.

Table 4: Existing DL distinguishers for members of reduced-round SPECK

Member	#R	Input Difference δ'	Output Mask λ'	Experiment	Ref.
SPECK32	9	(0211, 0a04)	(5820, 4020)	$2^{-8.93}$	[37]
	9	(a840, 0010)	(020a, 0204)	$2^{-7.30}$	[9]
	9	(000a, 0400)	(0e00, 0c00)	$2^{-11.58}$	[41]
	10	(0a20, 4205)	(5820, 4020)	$2^{-13.90}$	[37]
	10	(2050, 2040)	(3854, 3844)	$2^{-12.00}$	[9]
	10	(0211, 0a04)	(5820, 4020)	$2^{-11.58}$	[32]
	10	(2800, 0010)	(2300, 4380)	$2^{-2-10.2}$	[14]
	10	(8440, 8102)	(0e00, 0c00)	$2^{-14.58}$	[41]
	11	(2a10, 0004)	(3854, 3844)	$2^{-16.00}$	[9]
SPECK48	11	(0a20, 4205)	(7020, 6020)	$2^{-16.68}$	[32]
	11	(20082, 120200)	(8a805, 98804)	$2^{-17.55}$	[32]
SPECK64	11	(20082, 120200)	(c2801, d0800)	$2^{-5-6.27-3 \times 2}$	[14]
	11	(92400040, 40104200)	(4100400c, 4c004000)	$2^{-19.47}$	[32]
	12	(12029282, 02020282)	(61804028, 68004020)	—	[32]
SPECK96	13	(102490, 10801004)	(420c0200, 2400200)	$2^{-11-9.15-4 \times 2}$	[14]
	15	(20020028202, 120200049282)	(4d000203422b, c0002030223)	$2^{-20-5.62-8 \times 2}$	[14]
SPECK128	18	(40002403c012, 10020040000400c2)	(a49000000020343, 208000000020303)	$2^{-30-5.55-10 \times 2}$	[14]

C Re-compositions of DL Distinguisher for Speck32

Table 5: Hourglass(-like) re-decompositions of DL distinguishers of SPECK32

#R	Re-decompositions to hourglass(-like) structure				Experiment	Ref.
9	(0211, 0a04)	$\xrightarrow{2DC} (v_0[6])$	$\xrightarrow{5DL} (v_1[1])$	$\xrightarrow{2LC} (5820, 4020)$	$2^{-8.93}$	[37]
9	(a840, 0010)	$\xrightarrow{3DC} (v_1[12])$	$\xrightarrow{4DL} (v_1[5])$	$\xrightarrow{2LC} (020a, 0204)$	$2^{-7.30}$	[9]
9	(000a, 0400)	$\xrightarrow{1DC} (v_0[12])$	$\xrightarrow{7DL} (v_0[0])$	$\xrightarrow{1LC} (0e00, 0c00)$	$2^{-11.58}$	[41]
10	(0a20, 4205)	$\xrightarrow{3DC} (v_0[6])$	$\xrightarrow{5DL} (v_1[1])$	$\xrightarrow{2LC} (5820, 4020)$	$2^{-13.90}$	[37]
10	(2050, 2040)	$\xrightarrow{2DC} (v_1[10])$	$\xrightarrow{4DL} (v_1[3])$	$\xrightarrow{4LC} (3854, 3844)$	$2^{-12.00}$	[9]
10	(0211, 0a04)	$\xrightarrow{2DC} (v_0[6])$	$\xrightarrow{6DL} (v_1[1])$	$\xrightarrow{2LC} (5820, 4020)$	$2^{-11.58}$	[32]
10	(2800, 0010)	$\xrightarrow{1DC} (v_0[6])$	$\xrightarrow{6DL} (v_1[1])$	$\xrightarrow{3LC} (2300, 4380)$	$2^{-2-10.2}$	[14]
10	(8440, 8102)	$\xrightarrow{2DC} (v_0[12])$	$\xrightarrow{7DL} (v_0[0])$	$\xrightarrow{1LC} (0e00, 0c00)$	$2^{-14.58}$	[41]
11	(2a10, 0004)	$\xrightarrow{3DC} (v_1[10])$	$\xrightarrow{4DL} (v_1[3])$	$\xrightarrow{4LC} (3854, 3844)$	$2^{-16.00}$	[9]
11	(0a20, 4205)	$\xrightarrow{3DC} (v_0[6])$	$\xrightarrow{6DL} (v_1[1])$	$\xrightarrow{2LC} (7020, 6020)$	$2^{-16.68}$	[32]

D Re-compositions of DL Distinguisher for Reduced-round Alzette

Table 6: Existing DL distinguishers for reduced-round Alzette

Ref.	#R	Input Difference δ'	Output Mask λ'	Experiment
[37]	4	(80000000, 0)	(80000, 8)	1
[37]	5	(80000000, 0)	(80, 8000)	$2^{-0.13}$
[37]	6	(80000000, 0)	(40, 200000)	$2^{-1.45}$
[37]	8	(80020100, 10080)	(80000000, 8000)	$2^{-5.52}$
[32]	8	(80020100, 10080)	(c0010181, 01800001)	$2^{-4.069}$
[32]	9	(80020100, 10080)	(80600080, 60008000)	$2^{-7.60}$
[32]	10 [†]	(80020502, 10280)	(03810005, 80020180)	$2^{-10.52}$

†: [32] estimated the experimental DL correlation as $2^{-10.48}$ with 2^{30} samples, while we estimated it as $2^{-10.52}$ with 2^{35} samples.

Table 7: Hourglass(-like) re-decompositions of DL distinguishers of Alzette

Ref.	#R	Re-decompositions to hourglass(-like) structure		
[37]	4	$(v_0[31])$	$\xrightarrow{3DL} (v_1[3])$	$\xrightarrow{1LC} (80000, 8)$
[37]	5	$(v_0[31])$	$\xrightarrow{4DL} (v_1[15])$	$\xrightarrow{1LC} (80, 8000)$
[37]	6	$(v_0[31])$	$\xrightarrow{5DL} (v_1[21])$	$\xrightarrow{1LC} (40, 200000)$
[37]	8	(80020100, 10080)	$\xrightarrow{2DC} (v_1[16])$	$\xrightarrow{5DL} (v_1[15])$ $\xrightarrow{1LC} (80000000, 8000)$
[32]	8	(80020100, 10080)	$\xrightarrow{2DC} (v_1[16])$	$\xrightarrow{4DL} (v_0[0])$ $\xrightarrow{2LC} (c0010181, 01800001)$
[32]	9	(80020100, 10080)	$\xrightarrow{2DC} (v_1[16])$	$\xrightarrow{5DL} (v_1[15])$ $\xrightarrow{2LC} (80600080, 60008000)$
[32]	10	(80020502, 10280)	$\xrightarrow{2DC} (v_0[1])$	$\xrightarrow{6DL} (v_0[0])$ $\xrightarrow{2LC} (03810005, 80020180)$

E Re-decomposition of DL Distinguisher for Speck128, LEA and Cham

Speck128. The first 18-round DL distinguisher of SPECK128 is proposed in [14] shown as

$$\delta'_{opt} \xrightarrow[2^{-30}]{5DC} (v_0[53]) \xrightarrow[2^{-5.81}]{8DL} (v_0[13], v_1[5]) \xrightarrow[2^{-10}]{5LC} \lambda'_{opt},$$

with correlation estimated as $|bp\varepsilon^2| = 2^{-55.81}$. It can also be re-decomposed to the following form of (5, 9, 4)-hourglass structural DL distinguisher

$$\delta'_{opt} \xrightarrow[2^{-30}]{5DC} (v_0[53]) \xrightarrow{9DL} (v_1[5]) \xrightarrow[2^{-7}]{4LC} \lambda'_{opt}.$$

With 2^{34} pseudo-ciphertext pairs and 50 keys, we obtain the correlation of the middle 9-round DLA as $2^{-11.19}$. Thus the correlation can be re-estimated as $|bp\varepsilon^2| = 2^{-55.19}$.

LEA. In [14], the first DL attack on LEA is presented, where the following three DLAs ($\delta \xrightarrow{r_m DL} \lambda \xrightarrow{r_2 LC} \lambda'$) are used to construct 16- or 17-round DL distinguishers and to mount key recovery attacks.

- (1). $(v_3[26]) \xrightarrow{8DL} \underbrace{\sum \xrightarrow{3LC} (v_0[9], v_1[27], v_2[29], v_3[9, 0])}_{2^{-4.679}},$
- (2). $(v_3[31]) \xrightarrow{8DL} \underbrace{\sum \xrightarrow{4LC} (v_0[9], v_1[27], v_2[29], v_3[9, 0])}_{2^{-10.97}},$
- (3). $(v_3[31]) \xrightarrow[2^{-6.04}]{8DL} (v_1[10, 9], v_2[10, 9], v_3[8]) \xrightarrow[2^{-10}]{5LC} (v_0[18, 9, 6, 5], v_1[27, 24, 4], v_2[29, 6, 5], v_3[29, 0]).$

By placing one of the optimal 4-round DCs with probability 2^{-33} before DLA (2) and (3), 16- and 17-round DL distinguishers are obtained with absolute correlation estimated as $2^{-10.97} \times 2^{-33} = 2^{-43.97}$ and $2^{-6.04} \times 2^{-33-10 \times 2} = 2^{-59.04}$ respectively. We have deduced that the 2-round LC $(v_1[0]) \rightarrow (v_0[9], v_1[27], v_2[29], v_3[9, 0])$ has an absolute correlation 1. According to the linear hull, the term $\sum$ in DLA (1) includes those masks λ such that $\varepsilon(\lambda \xrightarrow{1LC} (v_1[0])) \cdot \mathcal{C}((v_3[26]) \xrightarrow{8DL} \lambda) \neq 0$, and the term $\sum$ in DLA (2) includes those masks λ such that $\varepsilon(\lambda \xrightarrow{2LC} (v_1[0])) \cdot \mathcal{C}((v_3[31]) \xrightarrow{8DL} \lambda) \neq 0$. Thus (1) and (2) can be re-decomposed to the following hourglass structural DLAs

$$\begin{aligned} (v_3[26]) &\xrightarrow{9DL} (v_1[0]) \xrightarrow[2^0=1]{2LC} (v_0[9], v_1[27], v_2[29], v_3[9, 0]), \\ (v_3[31]) &\xrightarrow{10DL} (v_1[0]) \xrightarrow[2^0=1]{2LC} (v_0[9], v_1[27], v_2[29], v_3[9, 0]) \end{aligned}$$

Similarly, DLA (3) can be re-decomposed to the following hourglass structural DLA

$$(v_3[31]) \xrightarrow{10DL} (v_1[0]) \xrightarrow[2^{-6}]{3LC} (v_0[18, 9, 6, 5], v_1[27, 24, 4], v_2[29, 6, 5], v_3[29, 0]).$$

In fact, we have found that DLA (3) can be obtained by extending DLA (2) forward by 1-round LC with absolute correlation 2^{-6} , thus the correlation can

be re-evaluated as $2^{-10.97} \times 2^{-6 \times 2} = 2^{-22.97}$, and thus the absolute correlation of the 17-round DL distinguisher is re-evaluated as $2^{-55.97}$, closer to the actual result than the original estimation $2^{-59.04}$ in [14].

Cham. Recently, building upon the MTIM search algorithm introduced in [14], [39] constructed a 46-round DL distinguisher of CHAM-64/128 with the correlation estimated as $|bp\varepsilon^2| = 2^{-31.08}$ and a 58-round one for CHAM-128/256 with $|bp\varepsilon^2| = 2^{-59.08}$, under $(r_1, r_m, r_2) = (6, 26, 14)$ and $(10, 32, 16)$ respectively, shown as follows

$$\begin{aligned} \delta'_{opt} \xrightarrow[2^{-10}]{6\text{DC}} (v_0[13]) &\xrightarrow[2^{-11.08}]{26\text{DL}} (v_0[0], v_1[15], v_2[10, 9], v_3[9, 8, 0]) \xrightarrow[2^{-5}]{14\text{LC}} \lambda'_{opt}, \\ \delta'_{opt} \xrightarrow[2^{-26}]{10\text{DC}} (v_0[27]) &\xrightarrow[2^{-15.08}]{32\text{DL}} (v_0[18], v_1[31, 30, 24, 17], v_2[23], v_3[15]) \xrightarrow[2^{-9}]{16\text{LC}} \lambda'_{opt}. \end{aligned}$$

We have proved that they can be re-decomposed to the $(6, 33, 7)$ - and $(10, 41, 7)$ -hourglass(-like) DL distinguishers as follows

$$\begin{aligned} \delta'_{opt} \xrightarrow[2^{-10}]{6\text{DC}} (v_0[13]) &\xrightarrow[2^{-3}]{33\text{DL}} (v_3[10]) \xrightarrow[2^{-3}]{7\text{LC}} \lambda'_{opt}, \\ \delta'_{opt} \xrightarrow[2^{-26}]{10\text{DC}} (v_0[27]) &\xrightarrow[2^{-3}]{41\text{DL}} (v_3[10, 9]) \xrightarrow[2^{-3}]{7\text{LC}} \lambda'_{opt}. \end{aligned}$$

F Support Experiments on Alzette

For Alzette, our experiments are done over 1-to-1, 1-to-2, 2-to-1 (δ, λ) pairs involving $\binom{64}{1} \times \binom{64}{1} + \binom{64}{1} \times \binom{64}{2} + \binom{64}{2} \times \binom{64}{1} = 262144$ possible (δ, λ) pairs. We construct 9- and 10-round DL distinguishers of Alzette under $(r_1, r_m, r_2) = (2, 4, 3)$ and $(2, 5, 3)$ respectively, and the experimental results are summarized in Table 8 and 9 respectively.

Table 8: Comparison of the number of hourglass(-like) and non-hourglass(-like) structural 9-round DL distinguishers under $(r_1, r_m, r_2) = (2, 4, 3)$

$-\log(bp\varepsilon^2)$	[7, 8]	[8, 9]	[9, 10]	[10, 11]
#hourglass(-like)	6	11	34	114
#non-hourglass(-like)	0	1	1	4
#total	6	12	35	118
ratio of hourglass(-like)	100%	92%	97%	97%

Specifically, we obtain three 10-round DL distinguishers with absolute correlations higher than 2^{-11} , and all of them can be re-decomposed to the hourglass structural ones. Among them, the best one is obtained as

$$\delta'_{opt} \xrightarrow[2^{-5}]{2\text{DC}} (v_0[1]) \xrightarrow[2^{-1}]{5\text{DL}} (v_0[0], v_1[24]) \xrightarrow[2^{-1}]{3\text{LC}} \lambda'_{opt}$$

which can be re-decomposed to $(2, 6, 2)$ -hourglass structural DL distinguisher as

$$\delta'_{opt} \xrightarrow[2^{-5}]{2\text{DC}} (v_0[1]) \xrightarrow[2^{-1}]{6\text{DL}} (v_0[0]) \xrightarrow[2^{-1}]{2\text{LC}} \lambda'_{opt},$$

Table 9: Comparison of the number of hourglass(-like) and non-hourglass(-like) structural 10-round DL distinguishers under $(r_1, r_m, r_2) = (2, 5, 3)$

$-\log(bp\varepsilon^2)$	[10, 11]	[11, 12]	[12, 13]	[13, 14]
#hourglass(-like)	3	2	8	19
#non-hourglass(-like)	0	0	0	4
#total	3	2	8	23
ratio of hourglass(-like)	100%	100%	100%	83%

exactly the re-decomposition to hourglass structure of the currently best 10-round DL distinguisher of Alzette in [32]. We also try the construction under $(r_1, r_m, r_2) = (3, 5, 2)$, and only obtain one with absolute correlation higher than 2^{-11} shown as

$$\delta'_{opt} \xrightarrow[2^{-6}]{3\text{DC}} (v_0[1], v_1[2]) \xrightarrow[5\text{DL}]{5\text{DL}} (v_0[0]) \xrightarrow[2^{-1}]{2\text{LC}} \lambda'_{opt},$$

which can be re-decomposed to the same hourglass structure as the currently best one.

G Algorithm for Preparing the SUBs Sets

Algorithm 6: Identify the SUBs sets for all 1-bit input differences

Input: A threshold ι ; Block size n ; The value of r_m ; A small positive integer t ; Sample size N' .

Output: SUBs sets $\mathcal{S}_\iota([i])$ for all $i = 0, \dots, n-1$.

// Prepare pseudo-ciphertext pairs for E_m^{-t}

```

1 for  $l = 1, \dots, N'$  do
2   Randomly generate the plaintext  $\mathbf{x}_l$ ;
3   Compute  $\mathbf{y}_l = E_m^{-t}(\mathbf{x}_l)$ , and  $\mathbf{y}_l^{[i]} = E_m^{-t}(\mathbf{x}_l \oplus [i])$  for  $i = 0, \dots, n-1$ ;
// Identify the SUBs sets  $\mathcal{S}_\iota([i])$  for  $(r_m - t)$ -round DLAs
4 for  $i = 0, \dots, n-1$  do
5    $\mathcal{S}_\iota([i]) \leftarrow []$ ;
6   for  $j = 0, \dots, n-1$  do
7     Compute the correlation  $|b_j^i|$  of  $[i] \xrightarrow{(r_m-t)\text{DL}} [j]$  over  $N'$  pairs  $(\mathbf{y}_l, \mathbf{y}_l^{[i]})$ ;
8     if  $|b_j^i| > \iota$  then
9       Add  $(j, |b_j^i|)$  to  $\mathcal{S}_\iota([i])$ ;
```

H An Example Illustrating the Two Concepts

As an example, to search for new 13-round DL distinguishers of SPECK64 ($cr = 6$) under $(r_1, r_m, r_2) = (3, 7, 3)$, we prepare the SUBs sets for 6-round DLAs stored in Speck64 6DL SUBs sets.txt by setting $\iota = 2^{-8}$. Given $\chi = 2^{-1}$,

we have found *collision* between $\delta = v_0[15]$ (at Round 3) and $\lambda = v_1[5]$ (at round 10) occurs at one multi-bit mask $\hat{\lambda}(v_0[13, 12], v_1[5, 2])$ (at Round 9). Looking up the row begin with “bit index: 47”, the absolute correlations of the 6-round DLAs under difference-mask pairs $((v_0[15]), (v_0[13])), ((v_0[15]), (v_0[12])), ((v_0[15]), (v_1[5])),$ and $((v_0[15]), (v_1[2]))$ are obtained as $2^{-5.00}$, $2^{-7.75}$, $2^{-6.48}$, and $2^{-6.25}$, respectively. The absolute correlation of the 7-round DLA $(v_0[15]) \rightarrow (v_1[5])$ is less than that of the 6-round one $(v_0[15]) \rightarrow (v_0[13, 12], v_1[5, 2])$, and thus less than $2^{-7.75}$ according to the Piling-up Lemma. Thus the *approximate weight* of $(v_0[15]) \rightarrow (v_1[5])$ is identified as 7.75.

I Improving DL Distinguishers of Speck96

We have constructed an improved DL distinguisher for 15-round SPECK96 under the decomposition parameters $(r_1, r_m, r_2) = (4, 7, 4)$. Besides, we construct the first 16-round DL distinguisher under the decomposition parameters $(r_1, r_m, r_2) = (4, 8, 4)$.

New 15-round DL distinguisher. First, we use Algorithm 2 with parameters $r_1 = 4$ and $r_2 = 4$ to prepare a total of $96 \cdot 190 \approx 2^{14.15}$ 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ stored in 19 separated tables $\mathcal{T}^{wt_0}, \dots, \mathcal{T}^{wt_{18}}$ with $wt_0 = 28, \dots, wt_{18} = 46$. Consequently, two 4-tuples are stored in $\mathcal{T}^{wt_0}$, 14 in $\mathcal{T}^{wt_1}$, 52 in $\mathcal{T}^{wt_2}, \dots$

Experiments suggest the critical-round number for DLAs of SPECK96 can take $cr = 6$. Then for the case that $r_m = 7(> cr)$, we follow the search procedure in Algorithm 4 to identify good candidates. Similar to the process of constructing 13-round DL distinguisher for SPECK64, we first use Algorithm 6 or Algorithm 5 (with the block size parameter $n = 96$) with parameters $\iota = 2^{-9}$, $r_m = 7$ and $t = 1$ to identify SUBs sets $\mathcal{S}_i(\delta)$ of 6-round DLAs for all 1-bit input differences, and then run Algorithm 4 with parameter $\chi = 2^{-1}$ to obtain 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ with high absolute correlations. Here we skip the detailed process and simply list some candidates of the 7-round DLAs as follows

$$\begin{aligned} & (v_1[43]) \xrightarrow{7DL} (v_1[10, 9]), (v_1[45]) \xrightarrow{7DL} (v_1[21]), (v_1[37]) \xrightarrow{7DL} (v_1[13]), \\ & (v_1[29]) \xrightarrow{7DL} (v_1[5]), (v_1[34]) \xrightarrow{7DL} (v_1[10]), (v_1[43]) \xrightarrow{7DL} (v_1[10]), (v_1[0]) \xrightarrow{7DL} (v_1[15]). \end{aligned}$$

Finally, we construct an improved 15-round DL distinguisher as

$$\begin{aligned} & (80800124a084, 808080002080) \xrightarrow[2^{-17}]{4DC} (v_1[43]) \xrightarrow[-2^{-9.36}]{7DL} (v_1[10, 9]) \\ & \xrightarrow[2^{-6}]{4LC} (4d2000604841, 410000604040), \end{aligned}$$

where the 7-round DLA is experimentally evaluated using 2^{35} pseudo-ciphertext pairs and 100 keys as $-2^{-9.36}$, and the total absolute correlation is estimated as $|bpe^2| = 2^{-38.36}$, improving the previous best result in [14] with absolute correlation $2^{-41.72}$.

The first 16-round DL distinguisher. With the prepared tables and SUBs sets for 6-round DLAs, we use Algorithm 4 with parameters $r_m = 8$, $t = 2$ and $\chi = 2^{-4}$ to construct good 8-round DLAs. The 16-round DL distinguisher is

constructed as

$$(800a080808, 800124a0848) \xrightarrow[2^{-22}]{4DC} (v_0[7]) \xrightarrow[2^{-10.32}]{8DL} (v_1[26]) \\ \xrightarrow[2^{-8}]{4LC} (404c41cd20, 4040404100).$$

J Improved DL Distinguishers for Reduced-round Alzette

We have improved all existing DL distinguishers over those in [32,37] for different reduced-round Alzette. In Table 10, we list our new and improved DL distinguishers for 4- to 7-round Alzette, while in Table 11 we list those for 8- to 14-round ones. In particular, we have constructed the first 11- to 14-round DL distinguishers, at most 4 rounds longer than the previous best ones.

Table 10: New and improved DL distinguishers for 4- to 7-round Alzette

#R	$\delta' \xrightarrow[p]{r_1 DC} \delta \xrightarrow[b]{r_m DL} \lambda \xrightarrow[\epsilon]{r_2 LC} \lambda'$	Experiment	Ref.
4	$(v_0[31]) \xrightarrow[2^0]{3DL} (v_1[3]) \xrightarrow[2^0]{1LC} (80000, 8)$	1	[37]
4	$(v_0[31]) \xrightarrow[2^0]{3DL} (v_1[a]) \xrightarrow[2^0]{1LC} (v_0[a+16], v_1[a]), \quad a = 1, 2, 3, 4, 5$	1	New
4	$(v_0[31]) \xrightarrow[2^0]{3DL} (v_0[0]) \xrightarrow[2^0]{1LC} (v_0[8], v_1[26])$	1	New
5	$(v_0[31]) \xrightarrow[2^{-0.13}]{4DL} (v_1[15]) \xrightarrow[2^0]{1LC} (80, 8000)$	$2^{-0.13}$	[37]
5	$(v_0[31], v_1[30]) \xrightarrow[2^0]{1DC} (v_1[30]) \xrightarrow[2^{-0.099}]{3DL} (v_1[6]) \xrightarrow[2^0]{1LC} (v_0[30], v_1[6])$	$2^{-0.099}$	New
6	$(v_0[31]) \xrightarrow[2^{-1.45}]{5DL} (v_1[21]) \xrightarrow[2^0]{1LC} (v_0[6], v_1[21])$	$2^{-1.448}$	[37]
6	$(v_0[31]) \xrightarrow[2^{-1.06}]{5DL} (v_1[29]) \xrightarrow[2^0]{1LC} (v_0[14], v_1[29])$	$2^{-1.057}$	New
6	$(v_0[31]) \xrightarrow[2^{-1.06}]{5DL} (v_1[29, 28]) \xrightarrow[2^0]{1LC} (v_0[14, 13], v_1[29, 28])$	$2^{-1.057}$	New
7	$(v_0[20], v_1[19]) \xrightarrow[2^{-1}]{1DC} (v_1[19]) \xrightarrow[2^{-1.34}]{5DL} (v_0[0]) \xrightarrow[2^0]{1LC} (v_0[31, 0], v_1[0])$	$2^{-2.333}$	New

J.1 Construction of 11-round DL Distinguisher

We specify the decomposition parameters as $(r_1, r_m, r_2) = (2, 6, 3)$ to construct 11-round DL distinguishers. First, we run Algorithm 2 with the 3-round LCs (round 8-11), to prepare tables storing $64 \times 126 \approx 2^{13}$ 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$. As a result, 18 tables $\mathcal{T}^{wt_0}, \mathcal{T}^{wt_1}, \dots, \mathcal{T}^{wt_{17}}$ are obtained, where $wt_0 = 6, wt_1 = 7, \dots, wt_{16} = 22, wt_{17} = 23$, with one 4-tuple stored in $\mathcal{T}^{wt_0}$, two in $\mathcal{T}^{wt_1}$, 42 in $\mathcal{T}^{wt_2}, \dots$, 1458 in $\mathcal{T}^{wt_{17}}$. We argue that the “critical-round number” for DLAs of Alzette can take $cr = 6$. Thus, we run Algorithm 3 to evaluate the 6-round DLAs by using 2^{32} pseudo-ciphertext pairs and 100 keys. Finally, we construct the first 11-round DL distinguisher as

$$(80020100, 10080) \xrightarrow[2^{-2}]{2DC} (v_1[16]) \xrightarrow[-2^{-6.53}]{6DL} (v_1[6, 5]) \xrightarrow[2^{-3}]{3LC} (c0a04031, 40c08061),$$

Table 11: New DL distinguishers for 8- to 14-round Alzette

Ref.	#R	$\delta' \xrightarrow[p]{r_1 \text{DC}} \delta \xrightarrow[b]{r_m \text{DL}} \lambda \xrightarrow[\epsilon]{r_2 \text{LC}} \lambda'$	$ bp\epsilon^2 $
[37]	8	$(80020100, 10080) \xrightarrow[2^{-2}]{2\text{DC}} (v_1[16]) \xrightarrow{5\text{DL}} (v_1[15]) \xrightarrow[2^0]{1\text{LC}} (80000000, 8000)$	$2^{-5.52}$
[32]	8	$(80020100, 10080) \xrightarrow[2^{-2}]{2\text{DC}} (v_1[16]) \xrightarrow{4\text{DL}} (v_0[0]) \xrightarrow[2^{-1}]{2\text{LC}} (c0010181, 01800001)$	$2^{-4.069}$
New	8	$(v_0[14], v_1[13]) \xrightarrow[2^{-1}]{1\text{DC}} (v_1[13]) \xrightarrow{5\text{DL}} (v_1[1]) \xrightarrow[2^0]{2\text{LC}} (v_0[17, 8, 0], v_1[24, 1])$	$2^{-4.058}$
[32]	9	$(80020100, 10080) \xrightarrow[2^{-2}]{2\text{DC}} (v_1[16]) \xrightarrow{5\text{DL}} (v_1[15]) \xrightarrow[2^{-1}]{2\text{LC}} (80600080, 60008000)$	$2^{-7.60}$
New	9	$(04020008, 02000004) \xrightarrow[2^{-4}]{2\text{DC}} (v_1[2]) \xrightarrow{5\text{DL}} (v_0[0]) \xrightarrow[2^{-1}]{2\text{LC}} (80810181, 81000080)$	$2^{-7.27}$
[32]	10	$(80020502, 10280) \xrightarrow[2^{-5}]{2\text{DC}} (v_0[1]) \xrightarrow{6\text{DL}} (v_0[0]) \xrightarrow[2^{-1}]{2\text{LC}} (03810005, 80020180)$	$2^{-10.52}$
New	10	$(80020100, 10080) \xrightarrow[2^{-2}]{2\text{DC}} (v_1[16]) \xrightarrow{6\text{DL}} (v_1[6, 5]) \xrightarrow[2^{-1}]{2\text{LC}} (40c00001, 8060)$	$2^{-10.53}$
New	11	$(80020100, 10080) \xrightarrow[2^{-2}]{2\text{DC}} (v_1[16]) \xrightarrow[2^{-6.53}]{6\text{DL}} (v_1[6, 5]) \xrightarrow[2^{-3}]{3\text{LC}} (c0a04031, 40c08061)$	$2^{-14.53}$
New	12	$(04040000, 06000000) \xrightarrow[2^{-4}]{2\text{DC}} (v_1[25]) \xrightarrow[2^{-11.02}]{7\text{DL}} (v_1[15, 14]) \xrightarrow[2^{-2}]{3\text{LC}} (004161c1, 01c08041)$	$2^{-19.02}$
New	13	$(a8400050, 40201028) \xrightarrow[2^{-11}]{3\text{DC}} (v_0[29]) \xrightarrow[2^{-9.32}]{7\text{DL}} (v_1[1]) \xrightarrow[2^{-2}]{3\text{LC}} (c2820001, 810100c2)$	$2^{-24.32}$
New	14	$(a8400050, 40201028) \xrightarrow[2^{-11}]{3\text{DC}} (v_0[29]) \xrightarrow[2^{-9.32}]{7\text{DL}} (v_1[1]) \xrightarrow[2^{-6}]{4\text{LC}} (8b0a0205, 81020484)$	$2^{-31.26}$

where the middle 6-round DLA is experimentally evaluated as $-2^{-6.53}$, thus the total absolute correlation can be estimated as $|bp\epsilon^2| = 2^{-14.53}$.

J.2 Construction of 12-round DL Distinguisher

We have tried many combinations of decomposition parameters (r_1, r_m, r_2) s.t. $r_1 + r_m + r_2 = 12$. As a result, the best choice is $r_1 = 2$, $r_m = 7$ and $r_2 = 3$, i.e., 2-round DC followed by 7-round DL trail and then followed by 3-round LC. After running Algorithm 2 with the 3-round LCs (round 9-12), we obtain 12 tables $\mathcal{T}^{wt_0}, \mathcal{T}^{wt_1}, \dots, \mathcal{T}^{wt_{11}}$, where $wt_0 = 6$, $wt_1 = 7, \dots, wt_{11} = 17$, with three 4-tuples stored in $\mathcal{T}^{wt_0}$, 6 in $\mathcal{T}^{wt_1}$, 147 in $\mathcal{T}^{wt_2}, \dots, 1593$ in $\mathcal{T}^{wt_{11}}$. Then for the case that $r_m = 7 (> cr)$, we use Algorithm 4 with the parameters $(t, \iota, \chi) = (1, 2^{-9}, 2^{-1})$ to find qualified 4-tuples $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$. Specifically, we set an absolute correlation threshold $\iota = 2^{-9}$ to identify the set of SUBs $\mathcal{S}_\iota(\delta)$ of 6-round DLAs (round 2-8) for each 1-bit input difference $\delta = v_0[i]$ and $\delta = v_1[i]$, $i = 0, \dots, 31$, by using $N' = 2^{31}$ ciphertext pairs. After that, we run Algorithm 4 with $\chi = 2^{-1}$ to obtain $(\delta'_{opt}, \delta, \lambda, \lambda'_{opt})$ with high absolute correlations.

During the process, some of the candidates of the 7-round DLAs are identified as $(v_1[1])$ or $(v_1[9])$ or $(v_1[17])$ or $(v_1[25]) \xrightarrow{7\text{DL}} (v_1[15])$ or $(v_1[15, 14])$. We use $N = 2^{38}$ ciphertext pairs to experimentally evaluate them and finally obtain the following 12-round DL distinguisher with $|bp\epsilon^2| = 2^{-19.02}$, shown as

$$(04040000, 06000000) \xrightarrow[2^{-4}]{2\text{DC}} (v_1[25]) \xrightarrow[2^{-11.02}]{7\text{DL}} (v_1[15, 14]) \xrightarrow[2^{-2}]{3\text{LC}} (c041c101, 8100c041).$$

J.3 Constructions of 13- and 14-round DL Distinguishers

We try many combinations of decomposition parameters (r_1, r_m, r_2) s.t. $r_1 + r_m + r_2 = 13$ to construct 13-round DL distinguishers. As a result, our best one is obtained with $|bp\varepsilon^2| = 2^{-24.32}$ shown as

$$(\text{a8400050}, 40201028) \xrightarrow[2^{-11}]{3\text{DC}} (v_0[29]) \xrightarrow[2^{-9.32}]{7\text{DL}} (v_1[1]) \xrightarrow[2^{-2}]{3\text{LC}} (\text{c2820001}, 810100\text{c2}).$$

Based on the same DLA, we also construct a $(3, 7, 4)$ -hourglass(-like) structural DL distinguisher for 14-round Alzette under the 4-round LC $(v_1[1]) \xrightarrow[2^{-6}]{4\text{LC}}$ $(8\text{b0a0205}, 81020484)$ (round 10-14). For tighter bound, $|\varepsilon|$ is experimentally confirmed as $2^{-5.47}$, thus we derive a better estimation of the total correlation as $2^{-31.26}$.

K Improved DL Distinguishers for SipHash

In [37], 3- and 4-round DL distinguishers are obtained with experimental correlations $2^{-0.78}$ and $2^{-6.03}$ respectively, as shown in the 1st and 3rd row of Table 12. Using our new search method, we construct significantly improved 3- and 4-round DL distinguishers, including deterministic ones with correlation ± 1 .

Table 12: Comparisons of DL distinguishers for the finalization of SipHash-2-4

Ref.	#R	Input Difference δ'	Output Mask λ'	Correlation
[37]	3	$(v_0[63])$	$(v_0[46, 14], v_1[46, 14], v_2[46, 14], v_3[46, 14])$	$2^{-0.78}$
New	3	$(v_2[63])$	$(v_1[18], v_2[50])$	1
[37]	4	$(v_0[31, 18], v_1[18])$	$(v_0[61, 29], v_1[61, 29], v_2[61, 29], v_3[61, 29])$	$2^{-6.03}$
New	4	$(v_2[6])$	$(v_0[21, 0], v_1[49], v_2[17], v_3[21])$	$2^{-0.13}$

K.1 Deterministic 3-round DL Distinguishers

We have constructed many *deterministic* 3-round DL distinguishers by choosing $r_1 = 0$, $r_m = 2.5$ and $r_2 = 0.5$. First, we run Algorithm 2 with $r_1 = 0$ and $r_2 = 0.5$. That is, we search the optimal 0.5-round LCs $\lambda \xrightarrow{0.5\text{LC}} \lambda'$ starting at round 2.5 for each 1-bit or consecutive 2-bit input mask λ , i.e., $\lambda = v_k[j]$ or $v_k[j+1, j]$ for $k = 0, 1, 2, 3$. After that, we find 256 λ yielding 0.5-round LCs with absolute correlation $2^0 = 1$, which are $v_0[0], v_2[0], v_1[j], v_3[j]$ for $j = 0, \dots, 63$ and $v_1[j+1, j], v_3[j+1, j]$ for $j = 0, \dots, 62$. Then, we find a number of 2.5-round DLAs starting with 1-bit input differences at round 0 and ending with the above 256 masks, which also have experimental correlation $2^0 = 1$. Above all, we obtain many 3-round DL distinguishers with absolute correlation 1, and one of such constructions is

$$(v_2[63]) \xrightarrow[2^0]{2.5\text{DL}} (v_1[1]) \xrightarrow[2^0]{0.5\text{LC}} (v_1[18], v_2[50]).$$

K.2 New 4-round DL Distinguishers

The best 4-round DL distinguisher is constructed by choosing $r_1 = 0$, $r_m = 3$, and $r_2 = 1$. By running the automatic model, we find that the 1-round LC starting at round 3 with $v_1[19]$ and $v_3[48]$ as input masks have correlation 1. Besides, we experimentally evaluate the 3-round DLAs. Finally, we obtain a 4-round DL distinguisher as

$$(v_2[6]) \xrightarrow[2^{-0.13}]{3\text{DL}} (v_1[19]) \xrightarrow[2^0]{1\text{LC}} (v_0[21, 0], v_1[49], v_2[17], v_3[21]),$$

whose experimental correlation is 0.911, i.e., $2^{-0.13}$, significantly improving the previous best one in [37] with absolute correlation $2^{-6.03}$. In addition, we also obtain other DL distinguishers that are better than that in [37].

L Recap on DL Distinguishers of ChaCha

L.1 DL Distinguishers with 1-bit Input Differences

In recent years, several works presented improvements on DL attacks against ChaCha, such as [7, 15, 16, 17, 19], where many DLAs are proposed and further used to construct DL distinguishers by appending high-correlation LCs, all the involved 7-round DL distinguishers can be re-decomposed into the (0, 3.5, 3.5)-hourglass structural ones in form of

$$\delta \xrightarrow{3.5\text{DL}} \lambda \xrightarrow{3.5\text{LC}} \lambda',$$

where $\delta = v_p[i]$, $p \in \{12, 13, 14, 15\}$, $i = 0, \dots, 31$, and $\lambda = v_q[0]$, $q \in \{0, \dots, 15\}$. For example, those in [15] and [17] are constructed with $(\delta, \lambda) = (v_{14}[6], v_3[0])$ and $(v_{14}[6], v_8[0])$ with time/data complexities 2^{214} and 2^{207} respectively. Generally, using a 1-bit flip on the input is quite natural to minimize the difference spread. Besides, when λ has only a single active bit in the least significant bit, it allows the correlation of the LC with λ as the input mask to be higher.

L.2 DL Distinguishers with 2-bit Input Differences in [8]

Distinguishers based on 3.5-round DLA. While all previous attacks only used 1-bit flip on the input, at ToSC/FSE 2023, Bellini *et al.* [8] showed new DLAs with 2-bit differences at the beginning, leading to better DL distinguishers than the previous ones using 1-bit input differences. Specifically, they constructed a 7-round DL distinguisher as

$$\underbrace{(v_{14}[21, 9]) \xrightarrow[2^{-7}]{1\text{DC}} (v_2[17, 5], v_6[24, 8], v_{10}[5, 1], v_{14}[25, 1])}_{3.5\text{DL}} \xrightarrow[2^{-18.75}]{2.5\text{DL}} (v_{11}[0]) \xrightarrow[2^{-37}]{3.5\text{LC}} \mathcal{A}, \quad (9)$$

where the correlation of middle 2.5-round DL is experimentally verified using 2^{42} samples as $2^{-18.75}$, and thus the total absolute correlation can be estimated as $2^{-18.75} \times 2^{-7-37 \times 2} = 2^{-99.75}$. Among those optimal 3.5-round LCs starting with input mask $(v_{11}[0])$ with absolute theoretical correlation 2^{-37} , the one used

in [8] (i.e., $\mathcal{A}$) has the absolute experimental correlation $2^{-34.6}$, which is derived by dividing the 3.5-round LC (3.5th to 7th round) into two sub-trails (3.5th to 6.5th, 6.5th to 7th) and experimentally verifying them as $2^{-19.30}$ and $2^{-15.389}$ respectively. Assuming independence between these two sub-trails, an estimation is obtained for the 3.5-round LC correlation as approximately $2^{-34.6}$. Based on this, they constructed a distinguisher for 7-round ChaCha with the time and data complexity $2^{(18.75+7+34.6 \times 2) \times 2} = 2^{189.9}$.

Distinguishers based on 4-round DLA. Additionally, [8] presented a 2-round DLA with experimental correlation $2^{-30.15}$, based on which they constructed better 7- and 7.5-round DL distinguishers. By extending this 2-round DLA backward with a 1-round DC with probability 2^{-7} and forward with a 1-round LC with absolute correlation 2^{-2} , a 4-round DLA with single-bit output mask ($v_2[0]$) can be interpreted as

$$\begin{aligned} v_{15}[29, 9] &\xrightarrow[2^{-7}]{1\text{DC}} (v_3[25, 5], v_7[28, 12], v_{11}[25, 21], v_{15}[21, 13]) \\ &\xrightarrow[2^{-30.15}]{2\text{DL}} (v_2[4, 3, 0], v_7[20, 4, 0], v_8[20, 19], v_{13}[4]) \xrightarrow[2^{-2}]{1\text{LC}} (v_2[0]) \end{aligned}$$

Here we emphasize that the above 2-round DLA has such a small absolute correlation that it is non-experimentally verifiable directly unless some assumptions are involved. As done in [8], instead of experimentally evaluating the 2-round DLA under the mask $(v_2[4, 3, 0], v_7[20, 4, 0], v_8[20, 19], v_{13}[4])$, they divide it into four partitions under the masks $v_2[4, 3, 0]$, $v_7[20, 4, 0]$, $v_8[20, 19]$ and $v_{13}[4]$, and experimentally evaluating the DL correlation for each such mask. Finally, the correlation value $2^{-30.15}$ is derived by multiplying those ending with masks $v_2[4, 3, 0]$ and $(v_7[20, 4, 0], v_8[20, 19], v_{13}[4])$, which are $2^{-5.25}$ and $2^{-24.9}$ respectively. For verifying such a small DL correlation value of $2^{-24.9}$, up to 2^{55} samples and 32 GPUs are used together with other optimizations, accounting for a large amount of computational resources.

Using this 4-round DLA, they construct a 7-round distinguisher by appending a 3-round LC with absolute theoretical correlation 2^{-24} (under the pattern “4DL+3LC”) and a 7.5-round one (under the pattern “4DL+3.5LC”) by combining a 3.5-round LC with absolute theoretical correlation 2^{-46} (under the pattern “4DL+3.5LC”). Thus the total correlations can be estimated as $2^{-30.15} \times 2^{-7-26 \times 2} = 2^{-89.15}$ and $2^{-30.15} \times 2^{-7-48 \times 2} = 2^{-133.15}$ respectively. In [8], Bellini *et al.* conducted an experimental verification of the 4-round LC starting at round 3 and ending at round 7 with absolute theoretical correlation 2^{-26} , which produced a value of $2^{-23.15}$. They also experimentally verified the 0.5-round LC starting at round 7 and ending at round 7.5 with absolute theoretical correlation 2^{-22} , which produced a value of $2^{-21.16}$. Using these values for estimating the 7- and 7.5-round DL distinguishers, the time/data complexities are $2^{(30.15+7+23.15 \times 2) \times 2} = 2^{166.9}$ and $2^{(30.15+7+(23.15+21.16) \times 2) \times 2} = 2^{251.54}$ respectively.

M Some Mask Values for ChaCha, Salsa and Forró

$\mathcal{B} = (v_0[28, 23, 16, 12, 8, 6, 0], v_1[31, 16, 8, 0], v_2[0], v_3[19, 18, 16, 12, 11, 7], v_4[19, 18, 17, 14], v_5[31, 30, 23, 19, 7], v_6[31, 30, 27, 26, 23, 22, 13, 11, 10, 7, 6], v_7[31, 30, 27, 26, 22, 19, 11, 6, 2], v_8[31, 28, 10, 8, 7], v_9[16, 12, 0], v_{10}[31, 26, 25, 23, 20, 16, 15, 8, 6, 4, 3], v_{11}[31, 27, 23, 20, 19, 8, 7, 4, 3, 0], v_{12}[28, 24, 20, 16, 15, 12, 11, 7, 4], v_{13}[24, 16, 7, 0], v_{14}[26, 19, 18, 8], v_{15}[27, 26, 23, 20, 19, 14, 12, 11, 8, 3, 2]).$
 $\mathcal{D} = (v_0[28, 26, 20, 16, 12, 3, 0], v_1[28, 24, 8, 7, 4, 0], v_2[24, 16], v_3[26, 18, 16, 12, 8, 7], v_4[16, 15, 14, 12, 11], v_5[31, 30, 28, 19, 16, 15, 8, 7, 0], v_6[31, 27, 26, 23, 22, 16, 13, 11, 8, 7, 6, 0], v_7[31, 30, 27, 22, 19, 14, 11, 10, 6, 2, 0], v_8[31, 28, 12, 11, 8, 7], v_9[23, 16, 12, 0], v_{10}[31, 26, 20, 18, 16, 15, 8, 7, 6, 4], v_{11}[31, 27, 20, 8, 7, 4, 2, 0], v_{12}[20, 12, 8, 0], v_{13}[16, 8, 0], v_{14}[24, 23, 10, 3], v_{15}[28, 24, 19, 11, 10, 7, 4]).$
 $\mathcal{E}' = (v_0[4, 0], v_2[18, 14, 4], v_3[18, 14], v_4[7, 4, 1], v_5[22, 21, 19, 18], v_6[22, 17, 16], v_7[31, 4, 1], v_8[24, 14, 13, 11, 6], v_9[19, 6, 5], v_{10}[24, 10, 5], v_{11}[31, 11, 10, 5], v_{12}[26, 14, 12], v_{13}[31, 26, 21, 12, 7, 0], v_{14}[31, 25, 24, 21, 20, 13], v_{15}[25, 17, 16, 7]).$
 $\mathcal{G} = (v_0[30, 19, 17, 8, 3, 2, 0], v_1[6, 3, 2], v_2[27, 3, 0], v_3[8, 0], v_4[10, 5, 3, 2], v_5[22, 21, 20, 0], v_6[30, 28, 27, 19, 5, 4, 0], v_7[18, 3, 0], v_8[27, 10, 5, 0], v_9[8, 0], v_{10}[21, 20], v_{11}[29, 28, 27, 8, 5, 4, 3, 2], v_{12}[30, 27, 26, 22, 21, 19, 13, 8, 0], v_{13}[27, 0], v_{14}[27, 22, 17, 16, 0], v_{15}[17, 16, 11, 3, 0]).$

N Theoretical Evaluations of DL Distinguishers

In this part, we propose refined approach for theoretically evaluating correlations of DLAs, with which we provide tighter theoretical evaluations of our newly found DL distinguishers for SPECK, Alzette, SipHash, and Chaskey, and for the first time, present theoretical interpretations for three previously reported experimental DL distinguishers of Chaskey without theoretical explanations ever before.

N.1 Theoretical Evaluation of DL Correlation by Niu *et al.* [37]

To validate our newly found DL distinguishers, we should provide theoretical evaluations for them. For this, we instantaneously adopt the currently most efficient algorithms in [37] to evaluate the theoretical DL correlations. In [37], Niu *et al.* presented an efficient algorithm for computing DL correlation of modular addition. Based on the local properties of modular additions and Morawiecki's technique [34], they show how to efficiently compute the DL correlations of iterative ARX primitives.

To extend Morawiecki's technique to evaluate the correlations of arbitrary DLAs of a cipher, one must be able to compute the DL correlation (with an arbitrary output linear mask) of each building block $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ of the cipher being analyzed with the knowledge of $\Pr\{x_i \oplus x'_i\}$ for all $i = 0, 1, \dots, n-1$. In [37], the following assumption is used in each of the middle rounds:

Assumption 1 ([37]) *Let $(\mathbf{x}, \mathbf{x}')$, $(\mathbf{y}, \mathbf{y}')$ be input pairs of the modular addition A (resp. S) s.t. $\Pr\{x_i \oplus x'_i = 1\} = pa_i$ and $\Pr\{y_i \oplus y'_i = 1\} = pb_i$, $i = 0, 1, \dots, n-1$. In the DL attack of iterative ARX primitives, assume that the events $x_i \oplus x'_i = 1$ and $y_i \oplus y'_i = 1$ for $i, j = 0, 1, \dots, n-1$ are mutually independent at each round.*

For computing DL correlations of modular additions $A(\mathbf{x}, \mathbf{y}) = \mathbf{x} \boxplus \mathbf{y}$, the underlying idea is actually the bit-slice technique proposed in [33] and applied in [38, 25, 43]. The general step is to first divide the addition modulo 2^n into the addition modulo 2 by introducing carry bits, and then pre-compute and store some useful matrices, and finally compute the correlation by doing some matrix multiplications using these pre-computed matrices.

Lemma 1 ([37]). *For any given input difference (δ_0, δ_1) and output mask λ , the DL correlation of the modular addition $A(\cdot)$ can be efficiently computed within a complexity of roughly $n \times 4 \times 4$ matrix multiplications as follows*

$$\mathcal{C}_A((\delta_0, \delta_1) \rightarrow \lambda) = (1, 1, 1, 1) \prod_{i=n-1}^0 \mathbf{U}_{(\alpha_i, \beta_i, \lambda_i)}(1, 0, 0, 0)^T,$$

where $\mathbf{U}_{(\alpha_i, \beta_i, \lambda_i)}$ are matrices pre-computed by Algorithm 7.

Algorithm 7: Construction of the matrices $\mathbf{U}_{(\alpha, \beta, \lambda)}$

Output: All the matrices $\mathbf{U}_{(\alpha, \beta, \lambda)}$ for $\alpha, \beta, \lambda \in \mathbb{F}_2$

```

1 Prepare a  $4 \times 4$  matrix  $\mathbf{N}$ ;
2 for  $\alpha, \beta, \lambda \in \mathbb{F}_2$  do
3   Create a matrix  $\mathbf{U}_{(\alpha, \beta, \lambda)}$  of size  $4 \times 4$ ;
4   Initialize  $\mathbf{N}$  with zeros;
5   for  $c_0 \in \{0, 1\}, d_0 \in \{0, 1\}, (x, y) \in \mathbb{F}_2 \times \mathbb{F}_2$  do
6     compute  $r = \lambda \cdot (\alpha \oplus \beta \oplus c_0 \oplus d_0)$ ;
7     compute  $c_1 = \varphi(x, y, c_0)$ ;
8     compute  $d_1 = \varphi(x \oplus \alpha, y \oplus \beta, d_0)$ ;
9     if  $r = 0$  then
10       $\mathbf{N}[c_1 + 2d_1][c_0 + 2d_0] := \mathbf{N}[c_1 + 2d_1][c_0 + 2d_0] + 1$ 
11    else
12       $\mathbf{N}[c_1 + 2d_1][c_0 + 2d_0] := \mathbf{N}[c_1 + 2d_1][c_0 + 2d_0] - 1$ 
13    for  $c_0, d_0, c_1, d_1 \in \{0, 1\}$  do
14       $\mathbf{U}_{(\alpha, \beta, \lambda)}[c_1 + 2d_1][c_0 + 2d_0] := \mathbf{N}[c_1 + 2d_1][c_0 + 2d_0]/2^2$ 
```

In most ARX ciphers, the nonlinear permutation $S(\mathbf{x}, \mathbf{y}) = (\mathbf{x} \boxplus \mathbf{y}, \mathbf{y})$, $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^n$, is often used. The correlation of $(\delta_0, \delta_1) \rightarrow (\lambda, \gamma)$ can be efficiently computed as

$$\mathcal{C}_S((\delta_0, \delta_1) \rightarrow (\lambda, \gamma)) = (1, 1, 1, 1) \prod_{i=n-1}^0 (-1)^{\gamma_i \beta_i} \mathbf{U}_{(\alpha_i, \beta_i, \lambda_i)}(1, 0, 0, 0)^T. \quad (10)$$

For computing DL correlation of iterative ARX primitives, let $\theta_a(p) = ((1 - a) - (-1)^a p)$ for $a \in \mathbb{F}_2$, i.e., $\theta_0(p) = 1 - p$ and $\theta_1(p) = p$. Under Assumption 1, the following results are derived.

Lemma 2 ([37]). *For output linear mask $(\lambda, \gamma) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$, the DL correlation of $S(\mathbf{x}, \mathbf{y}) = (\mathbf{x} \boxplus \mathbf{y}, \mathbf{y})$ can be computed as*

$$\mathcal{C}_S(\lambda, \gamma) = (1, 1, 1, 1) \prod_{i=n-1}^0 \mathbf{H}_{(\lambda_i, \gamma_i)}^{p_{a_i}, p_{b_i}}(1, 0, 0, 0)^T,$$

where $\mathbf{H}_{(\lambda_i, \gamma_i)}^{pa_i, pb_i} = \sum_{a, b \in \mathbb{F}_2} (-1)^{\gamma_i b} \theta_a(pa_i) \theta_b(pb_i) \mathbf{U}_{(a, b, \lambda_i)}$ are 4×4 matrices.

Lemma 3 ([37]). *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be a vectorial Boolean function mapping $\mathbf{x} \in \mathbb{F}_2^n$ to $L \circ S(\mathbf{x}) \oplus \mathbf{c}$, where $\mathbf{c} \in \mathbb{F}_2^n$ is a constant, $S : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is a nonlinear permutation and L is an $n \times n$ binary matrix. The DL correlation of F with difference-mask pair (Δ, λ) can be computed as $C_F(\Delta, \lambda) = C_S(\Delta, L^T(\lambda))$.*

Using the above method, we compute the theoretical correlations of our newly found DL distinguishers. We have found, however, the theoretical results do not match well with the experimental results (see the last columns of Table 14 and Table 15. We try to figure out the reasons as follows:

- For a single round DLA with one modular addition involved, the correlation can be efficiently and accurately computed by combining Lemma 3 and Eq.(10), and even for very few rounds ones with very few modular additions and XORs, the theoretical correlations predicted by [37] will not deviate much from the actual results.
- In the case of (dependent) chains of modular additions and XORs, however, the accuracy of the theoretical estimation for DL correlation in [37] is affected by the independency assumption among any two bits in each of the middle rounds, which is usually hard to hold in practice. The errors will accumulate and may lead to large deviations between the predicted and actual results.

N.2 Towards Tighter Theoretical Evaluations of DL Correlations

In this part, we try to provide tighter theoretical evaluations in an attempt to narrow the gap between theory and experiments.

The underlying idea of our strategy is quite intuitive. For a given number of round of DLA of an ARX cipher, we attempt to extend the length of the DC part forward and that of the LC part backward as much as possible by using the efficient algorithms in [31,40], and then use the efficient algorithm in [37] to compute the DL correlation for the remaining short middle rounds, thus reducing the usage of independency assumptions in middle rounds. With the above, we hope to achieve a balance between the accuracy of the DL correlation and efficiency.

In our strategy, the target T -round DLA is suitably split into $T = d + m + l$ rounds, written as $H = H_l \circ H_m \circ H_d$, representing d -round DC followed by m -round DL then followed by l -round LC. According to the linear hull of DL correlation, we have

$$C_H(\delta \rightarrow \lambda) = \sum_{\delta^{(d)}} \sum_{\lambda^{(l)}} \Pr_{H_d}[\delta \rightarrow \delta^{(d)}] \cdot C_{H_m}(\delta^{(d)} \rightarrow \lambda^{(l)}) \cdot \varepsilon_{H_l^{-1}}^2(\lambda^{(l)} \leftarrow \lambda),$$

where $\delta^{(d)}$ s are all possible output differences of the effective d -round DCs, and $\lambda^{(l)}$ s are all possible input masks of the effective l -round LCs, as defined in Definition 2 and 1 respectively, and each $(\delta^{(d)}, \lambda^{(l)})$ pair constitutes an effective DL trail $\delta^{(d)} \rightarrow \lambda^{(l)}$.

Generally, there are a lot of $(\delta^{(d)}, \lambda^{(l)})$ pairs available. Considering the efficiency, we set the probability threshold for DC and the linear correlation threshold for LC. We use $\mathcal{D}(d, w^{(d)})$ or $\mathcal{D}$ to denote the cluster of d -round DCs (d ADDs involved) with probability $\geq 2^{-w^{(d)}}$, and use $\mathcal{L}(l, w^{(l)})$ or $\mathcal{L}$ the cluster of l -round LCs (l ADDs involved) with absolute linear correlation $\geq 2^{-w^{(l)}}$, i.e.,

$$\begin{aligned}\mathcal{D}(d, w^{(d)}) &= \{\tau_d = (\delta \rightarrow \delta^{(d)}) : \Pr_{H_d}[\tau_d] \geq 2^{-w^{(d)}}\} \\ \mathcal{L}(l, w^{(l)}) &= \{\tau_l = (\lambda^{(l)} \leftarrow \lambda) : |\varepsilon_{H_l^{-1}}[\tau_l]| \geq 2^{-w^{(l)}}\}\end{aligned}$$

then the T -round DL correlation can be approximated as

$$\mathcal{C}_H(\delta \rightarrow \lambda) \approx \sum_{\delta^{(d)} \in \mathcal{D}} \sum_{\lambda^{(l)} \in \mathcal{L}} \Pr_{H_d}[\delta \rightarrow \delta^{(d)}] \cdot \mathcal{C}_{H_m}(\delta^{(d)} \rightarrow \lambda^{(l)}) \cdot \varepsilon_{H_l^{-1}}^2(\lambda^{(l)} \leftarrow \lambda). \quad (11)$$

We will use the efficient algorithms in [31] and [40] to compute the probability of DC and linear correlation of LC respectively, while that in [37] to compute the DL correlation values $\mathcal{C}_{H_m}(\delta^{(d)} \rightarrow \lambda^{(l)})$ for $|\mathcal{D}| \times |\mathcal{L}|$ pairs of $(\delta^{(d)}, \lambda^{(l)})$.

To illustrate the effectiveness, we will provide theoretical evaluations of our newly found DL distinguishers SPECK, Alzette, and the underlying permutations of SipHash and Chaskey, and also give the first theoretical interpretations on the previous experimental DL distinguishers of Chaskey without theoretical explanations ever before.

N.3 Theoretical Evaluations of DL Correlations of Chaskey

In [7,30], the following three 4-round experimental DLAs are employed to mount attacks without theoretical justifications ever before.

DL APPL. 1. $(v_0[22])_{R_1} \rightarrow (v_0[16])_{R_5}$, with experimental correlation $|b| 2^{-6.1}$,

DL APPL. 2. $(v_0[31])_{R_{1.5}} \rightarrow (v_2[20])_{R_{5.5}}$, with experimental correlation $|b| 2^{-5.1}$,

DL APPL. 3. $(v_0[31])_{R_{1.5}} \rightarrow (v_2[20, 19])_{R_{5.5}}$, with experimental correlation $|b| 2^{-5.1}$.

By adopting the above strategy, we obtain the theoretical correlation of DL APPL. 1 as $\mathcal{C}_{th} = 2^{-11.55}$, under the split parameters $d = 1.5$, $m = 1$ and $l = 1.5$ and the cluster weights $w^{(d)} = 9$ and $w^{(l)} = 3$. The gap between our theoretical estimation and the experimental result may come from two aspects: (1) Due to the threshold, i.e., $w^{(d)}$ and $w^{(l)}$, lots of valid DCs and LCs are not included. Increasing the values of $w^{(d)}$ or $w^{(l)}$ will lead to theoretical evaluation closer to the experimental result, but will also decrease the efficiency. (2) Assumption 1 is used in the process of computing each of the 1-round DL correlations, which will affect the accuracy of the theoretical prediction. In any case, our theoretical result is an experimentally verifiable result, whose standard deviation assuming a normal distribution is $\mathcal{O}(1/\mathcal{C}_{th}^2)$, i.e., $2^{23.1}$. Similarly, the theoretical correlations of DL APPL. 2 and DL APPL. 3 are both derived as $2^{-10.96}$ under the split parameters $d = 2$, $m = 1$ and $l = 1$ and the cluster weights $w^{(d)} = 12$ and $w^{(l)} = 5$.

In the following, we present the detailed process for deducing the theoretical evaluations.

Theoretical DL Correlation Evaluation of DL APPL. 1. After the analysis, we split the target 4-round DL part into $1.5 + 1 + 1.5$ rounds, written as $H = H_{1.5} \circ H_1 \circ H_{1.5}$, representing 1.5-round DC followed by 1-round DL then followed by 1.5-round LC. Note here though $m = 1$, i.e., 1-round DL, two chains of modular additions involved, each with 2 ADDs.

After specifying the split values, we choose the cluster weights $w^{(d)}$ and $w^{(l)}$. For the DC part, the largest probability of the 1.5-round DC is 2^{-7} . For different $w^{(d)} (\geq 7)$, we have

- when $w^{(d)} = 7$, we obtain $|\mathcal{D}(d, w^{(d)})| = 1$ DCs.
- when $w^{(d)} = 8$, we obtain $|\mathcal{D}(d, w^{(d)})| = 5$ DCs.
- when $w^{(d)} = 9$, we obtain $|\mathcal{D}(d, w^{(d)})| = 9$ DCs.
- when $w^{(d)} = 10$, we obtain $|\mathcal{D}(d, w^{(d)})| = 20$ DCs.

As for the LC part, we note that the 1-round LC trail

$$(v_0[0], v_1[0], v_2[0], v_3[27, 0])_{R_4} \rightarrow (v_2[0], v_3[0])_{R_{4.5}} \rightarrow (v_0[16])_{R_5}$$

has the absolute correlation 1. Further extending reversely the LC part for 0.5-round, we obtain $\lambda^{(l)}$. The largest absolute correlation of the 1.5-round LC is 2^{-2} . For different $w^{(l)} (\geq 2)$, we have

- when $w^{(l)} = 2$, we obtain $|\mathcal{L}(l, w^{(l)})| = 4$ LCs.
- when $w^{(l)} = 3$, we obtain $|\mathcal{L}(l, w^{(l)})| = 20$ LCs.
- when $w^{(l)} = 4$, we obtain $|\mathcal{L}(l, w^{(l)})| = 36$ LCs.

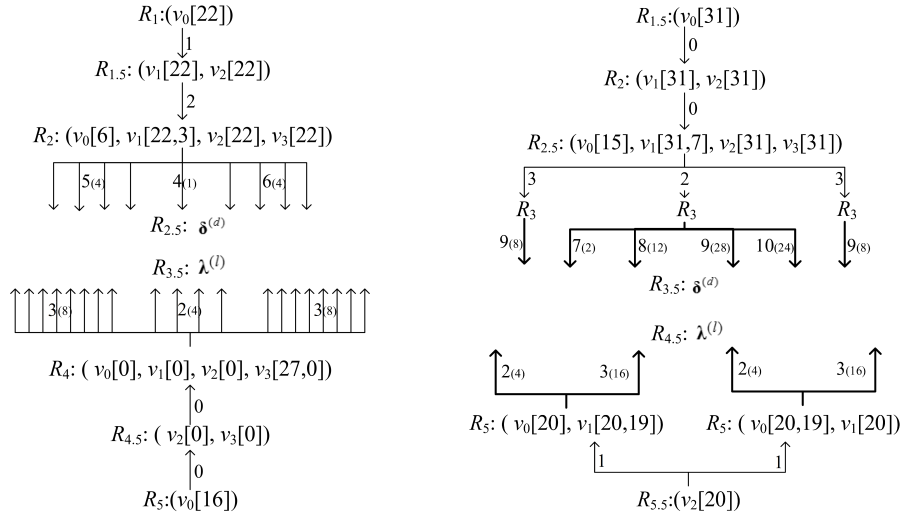


Fig. 12: On the left side, the DC and LC extending schemes of DL APPL. 1. On the right side, the DC and LC extending schemes of DL APPL. 2/3

On the left side of Fig. 12, we give our choices for the DC part extending and LC part reverse extending schemes, where $w^{(d)} = 9$ and $w^{(l)} = 3$. According

to Eq.(11), we need to compute $|\mathcal{D}| \times |\mathcal{L}| = 180$ 1-round DL correlations from round 2.5 to round 3.5. Combining the efficient algorithms in [37] as shown in Lemma 2 and Lemma 3, we obtain the theoretical correlation of DL APPL. 1 as $\mathcal{C}_{th} = 2^{-11.55}$.

Theoretical DL Correlation Evaluation of DL APPL. 2/3. DL APPL. 2 and DL APPL. 3 differ only in the output linear mask, thus the extending schemes of the DC part are the same, while the reverse extending schemes of the LC part are quite similar. For both of them, we split the target 4-round DL part into $2 + 1 + 1$ rounds, representing “2DC+1DL+1LC”. After specifying the split values, we choose the cluster weights $w^{(d)}$ and $w^{(l)}$.

For the 2-round DC, the largest probability is 2^{-9} . For the 1-round LC, the largest absolute correlation is 2^{-3} . For different $w^{(d)}(\geq 9)$ and $w^{(l)}(\geq 3)$, the values of $|\mathcal{D}(d, w^{(d)})|$ and $|\mathcal{L}(l, w^{(l)})|$ are shown as follows.

$\frac{w^{(d)}(\geq 9)}{ \mathcal{D}(d, w^{(d)}) }$	$\frac{9}{2}$	$\frac{10}{14}$	$\frac{11}{42}$	$\frac{12}{82}$
$\frac{w^{(l)}(\geq 3)}{ \mathcal{L}(l, w^{(l)}) }$	$\frac{3}{8}$	$\frac{4}{40}$	$\frac{5}{72}$	

For different choices of $(w^{(d)}, w^{(l)})$, the theoretical evaluations for DL APPL. 2 are listed in Table 13. All these theoretical results $\mathcal{C}_{th}$ in Table 13 are experimentally verifiable with standard deviation $\mathcal{O}(1/\mathcal{C}_{th}^2)$ by assuming a normal distribution. On the right side of Fig.12, we show one of our choices for the DC part extending and LC part reverse extending schemes, where $w^{(d)} = 12$ and $w^{(l)} = 4$. Though the LC schemes are a little bit different from that of DL APPL. 2. The theoretical evaluation for DL APPL. 3 is the same.

Table 13: Theoretical evaluations for DL APPL. 2/3

$(w^{(d)}, w^{(l)})$	$ \mathcal{D} \cdot \mathcal{L} $	$\mathcal{C}_{th}$
(10, 4)	560	$2^{-11.90}$
(11, 4)	1680	$2^{-11.33}$
(12, 4)	3280	$2^{-11.07}$
(12, 5)	5904	$2^{-10.96}$

N.4 Theoretical Evaluations of Newly Found DL Distinguishers

Here we provide the theoretical DL correlations of our newly found DLAs for SPECK, Alzette, SipHash and Chaskey, as shown in Table 14 and Table 15.

Table 14: Theoretical DL correlations for reduced-round SPECK

Member #R	DL APPL.	Correlation		
		Experiment	Our method* ($w^{(d)}, w^{(l)}$)	[37] [†]
SPECK32 11	$(v_1[3]) \xrightarrow{4DL} (v_1[3])$	$2^{-1.89}$	$2^{-2.28} (2DC+1DL+1LC, (7,3))$	$2^{-5.79}$
11	$(v_1[29]) \xrightarrow{5DL} (v_1[13])$	$2^{-1.82}$	$2^{-2.64} (2DC+2DL+1LC, (9,3))$	$2^{-6.88}$
SPECK64 12	$(v_1[29]) \xrightarrow{6DL} (v_1[5])$	$2^{-9.29}$	$2^{-11.98} (3DC+2DL+1LC, (14,3))$	$2^{-26.24}$
13	$(v_0[15]) \xrightarrow{7DL} (v_1[5])$	$2^{-9.59}$	$2^{-12.45} (4DC+2DL+1LC, (15,3))$	$2^{-64.00}$
15	$(v_1[43]) \xrightarrow{7DL} (v_1[10, 9])$	$2^{-9.36}$	$2^{-13.98} (3DC+2DL+2LC, (15,6))$	$2^{-42.98}$
SPECK96 16	$(v_0[7]) \xrightarrow{8DL} (v_1[26])$	$2^{-10.32}$	$2^{-15.63} (4DC+2DL+2LC, (15,6))$	$2^{-94.04}$

*: Results derived by our theoretical evaluation methods for corresponding patterns.

†: Results derived by Niu *et al.*'s method in [37] for corresponding patterns.

Table 15: Theoretical DL correlations for reduced-round Alzette, SipHash and Chaskey

Ciphers	#R	DL APPL.	Correlation		
			Experiment	Our method ($w^{(d)}, w^{(l)}$)	[37]
Alzette	9	$(v_1[2])_{R_2} \xrightarrow{5DL} (v_0[0])_{R_7}$	$2^{-1.24}$	$2^{-1.66} (2DC+1DL+2LC, (7,3))$	$2^{-6.04}$
	10/11	$(v_1[16])_{R_2} \xrightarrow{6DL} (v_1[6, 5])_{R_8}$	$2^{-6.53}$	$2^{-9.10} (3DC+2DL+1LC, (14,3))$	$2^{-106.69}$
	12	$(v_1[25])_{R_2} \xrightarrow{7DL} (v_1[15, 14])_{R_9}$	$2^{-11.02}$	$2^{-13.92} (3DC+2DL+2LC, (14,6))$	$2^{-\infty}$
	13/14	$(v_0[29])_{R_3} \xrightarrow{7DL} (v_1[1])_{R_{10}}$	$2^{-9.32}$	$2^{-12.86} (3DC+2DL+2LC, (14,6))$	$2^{-\infty}$
SipHash-2-4	3	$(v_2[63])_{R_0} \xrightarrow{2.5DL} (v_1[1])_{R_{2.5}}$	1	0.999981	0.999981
	4	$(v_2[6])_{R_0} \xrightarrow{3DL} (v_1[19])_{R_3}$	$2^{-0.13}$	$2^{-6.06} (0.5DC+2DL+0.5LC, (1,1))$	$2^{-32.26}$
Chaskey	7	$(v_0[31])_{R_{1.5}} \xrightarrow{4DL} (v_3[11])_{R_{5.5}}$	$2^{-8.65}$	$2^{-17.47} (2DC+1DL+1LC, (12,5))$	$2^{-59.35}$